

2.1

SUPPLY AND DEMAND I: HOW MARKETS WORK





2.1

The Market Forces of Supply and Demand

- Supply and demand are the two words that economists use most often.
- Supply and demand are the forces that make market economies work.
- Modern microeconomics is about supply, demand, and market equilibrium.

MARKETS AND COMPETITION

- A *market* is a group of buyers and sellers of a particular good or service.



- The terms supply and demand refer to the behavior of people . . . as they interact with one another in markets.

MARKETS AND COMPETITION

- The buyers as a group determine the demand for the product



- the sellers as a group determine the supply of the product

Competitive Markets

- A *competitive market* is a market in which there are many buyers and sellers so that each has a negligible impact on the market price.

Competition: Perfect and Otherwise

- Perfect Competition
 - Products are the same
 - Numerous buyers and sellers so that each has no influence over price
 - Price and quantity are determined by all buyers and sellers
 - As they interact in the marketplace
 - Buyers and Sellers are price takers
- Monopoly
 - One seller, and seller controls price

Competition: Perfect and Otherwise

- Oligopoly
 - Few sellers
 - Not always aggressive competition
- Monopolistic Competition
 - Many sellers
 - Slightly differentiated products
 - Each seller may set price for its own product

DEMAND

- *Quantity demanded* is the amount of a good that buyers are willing and able to purchase.
- Law of Demand
 - The *law of demand* states that, other things equal, the quantity demanded of a good falls when the price of the good rises.

Demand

- Demand schedule - a table
 - The *demand schedule* is a table that shows the relationship between the price of the good and the quantity demanded.
- Demand curve - a graph
 - Relationship between the price of a good and quantity demanded
- Individual demand
 - Demand of one individual

Catherine's Demand Schedule

**Price of
Ice-Cream Cone**

**Quantity of
Cones Demanded**



\$0.00

12

0.50

10

1.00

8

1.50

6

2.00

4

2.50

2

3.00

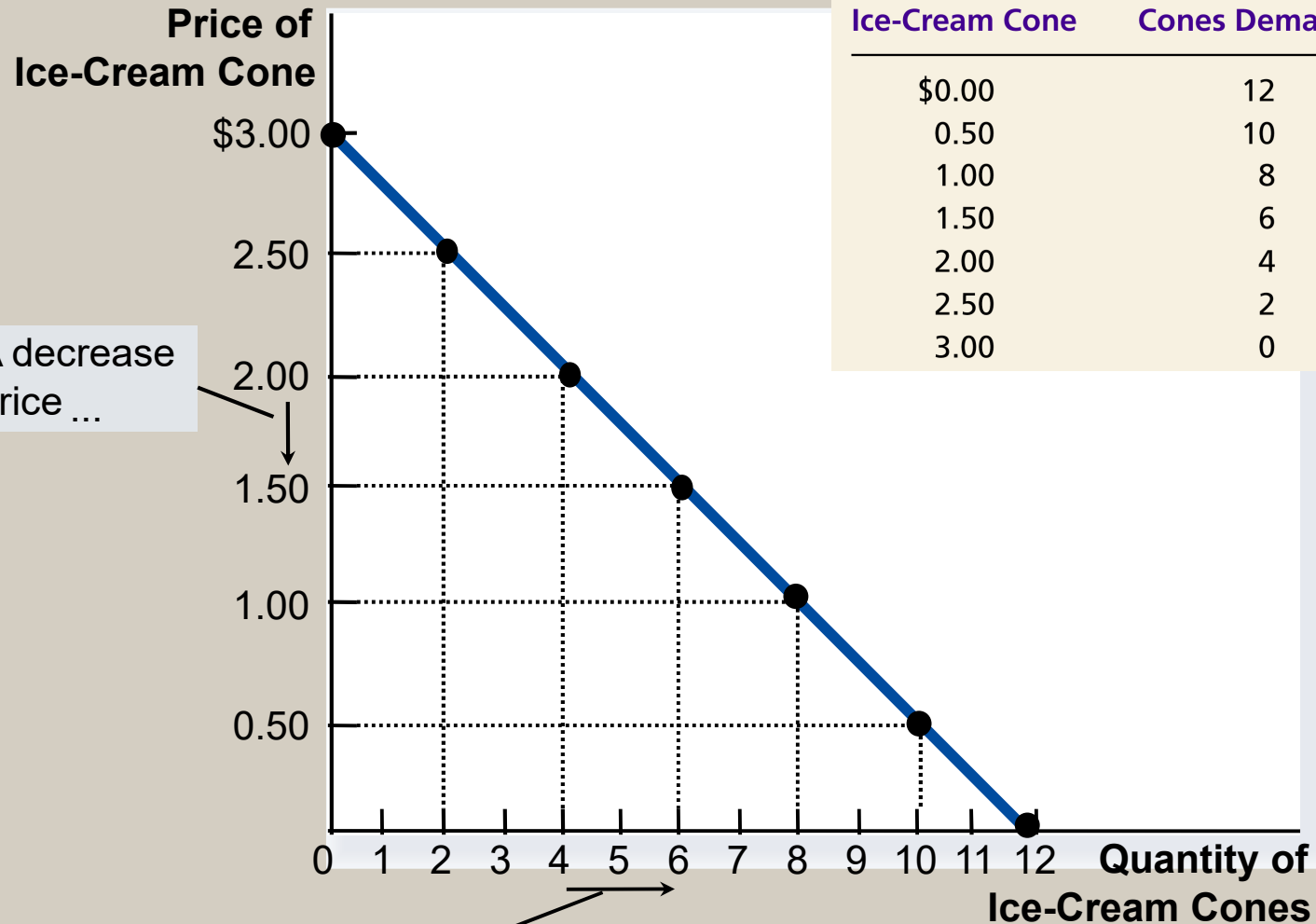
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The Demand Curve: The Relationship between Price and Quantity Demanded

- Demand Curve
 - The *demand curve* is a graph of the relationship between the price of a good and the quantity demanded.

Figure 1 Catherine's Demand Schedule and Demand Curve



Price of Ice-Cream Cone	Quantity of Cones Demanded
\$0.00	12
0.50	10
1.00	8
1.50	6
2.00	4
2.50	2
3.00	0

Market Demand versus Individual Demand

- Market demand refers to the sum of all individual demands for a particular good or service.
- Graphically, individual demand curves are summed horizontally to obtain the market demand curve.

Market Demand as the Sum of Individual Demands

Catherine's demand

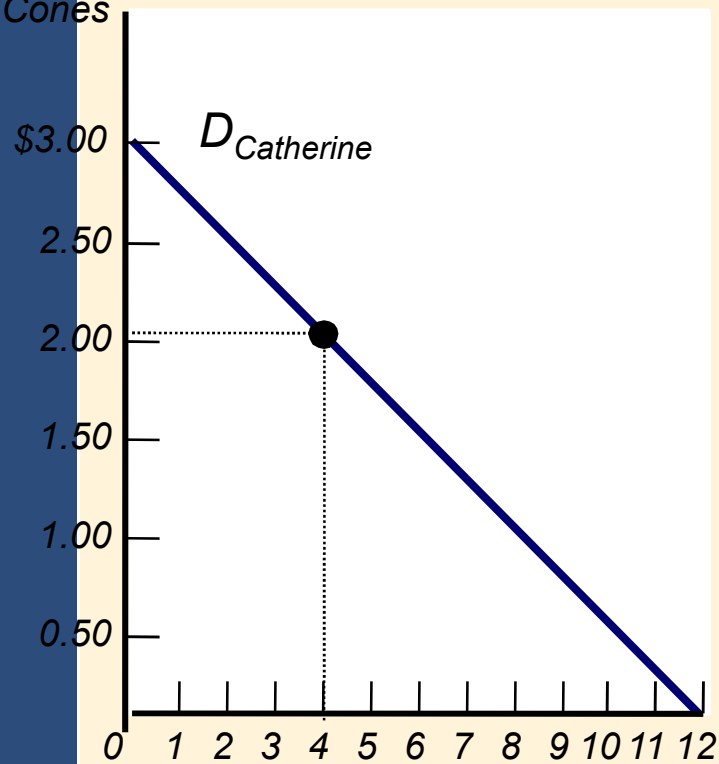
+

Nicholas's demand

=

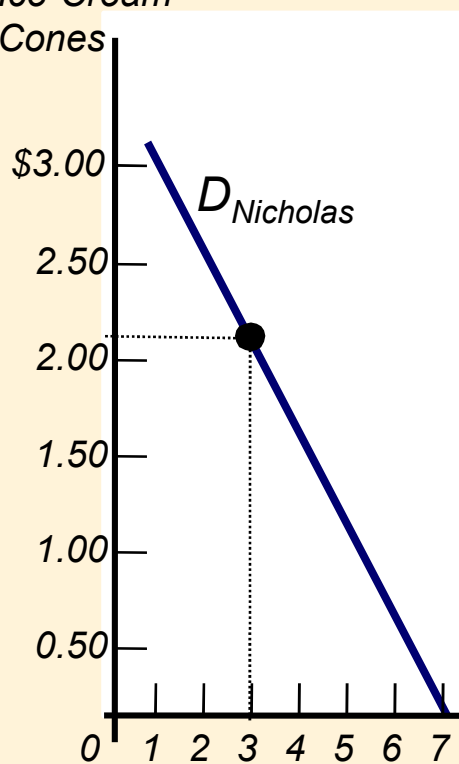
Market demand

Price of
Ice-Cream
Cones



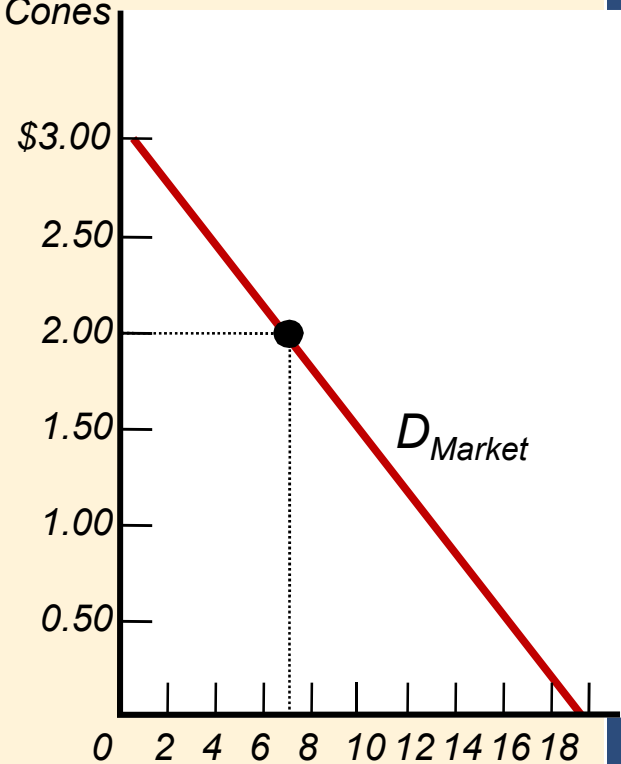
Quantity of Ice-Cream Cones

Price of
Ice-Cream
Cones



Quantity of Ice-Cream Cones

Price of
Ice-Cream
Cones



Quantity of Ice-Cream Cones

Example

Suppose the individual demand function of a product is given by: $P = 10 - Q / 2$

and there are about 100 identical buyers in the market.

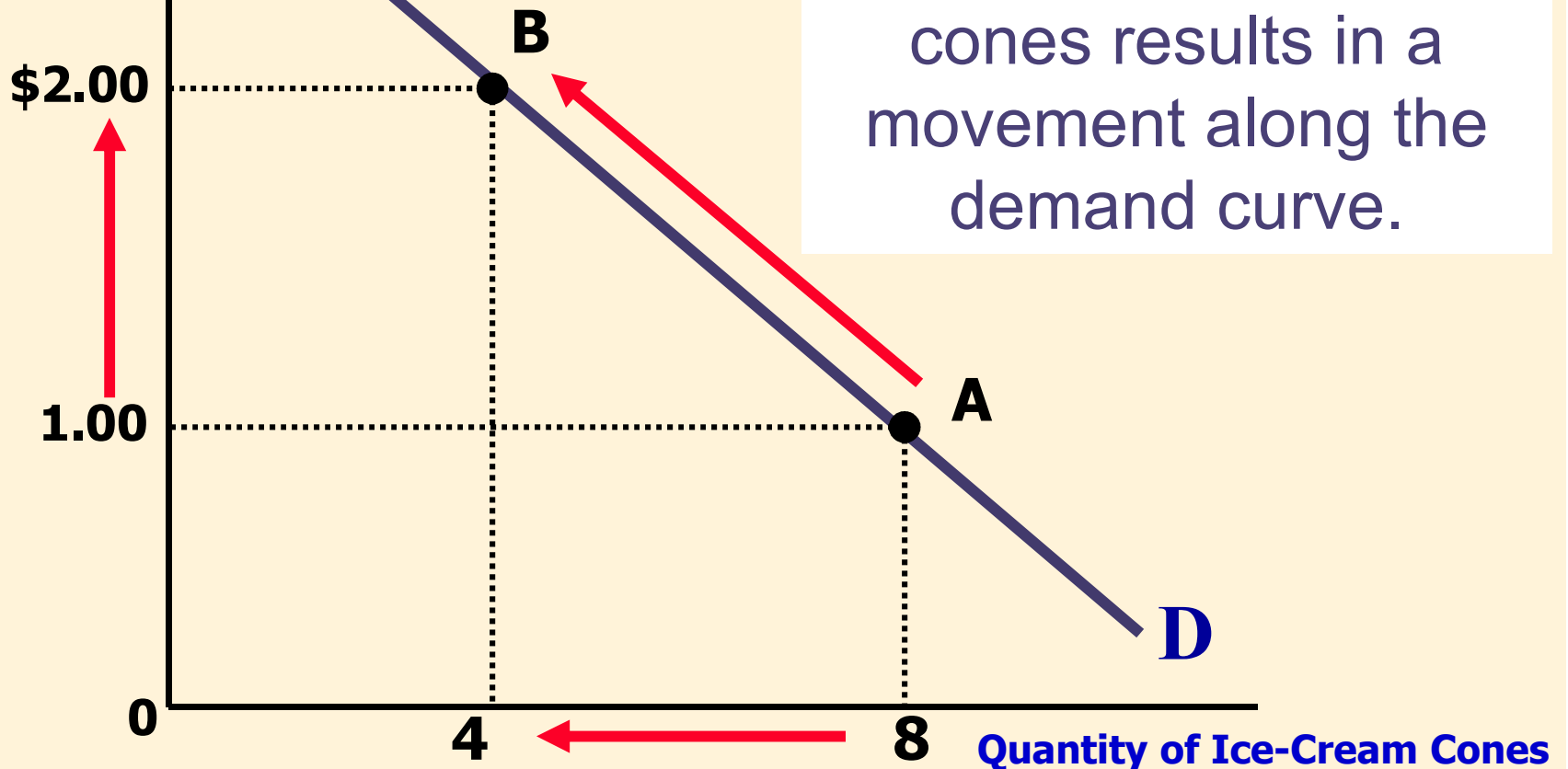
Derive the market demand curve:

Shifts in the Demand Curve

- Change in Quantity Demanded
 - Movement along the demand curve.
 - Caused by a change in the price of the product.

Changes in Quantity Demanded

Price of Ice-Cream Cones



Shifts in the Demand Curve

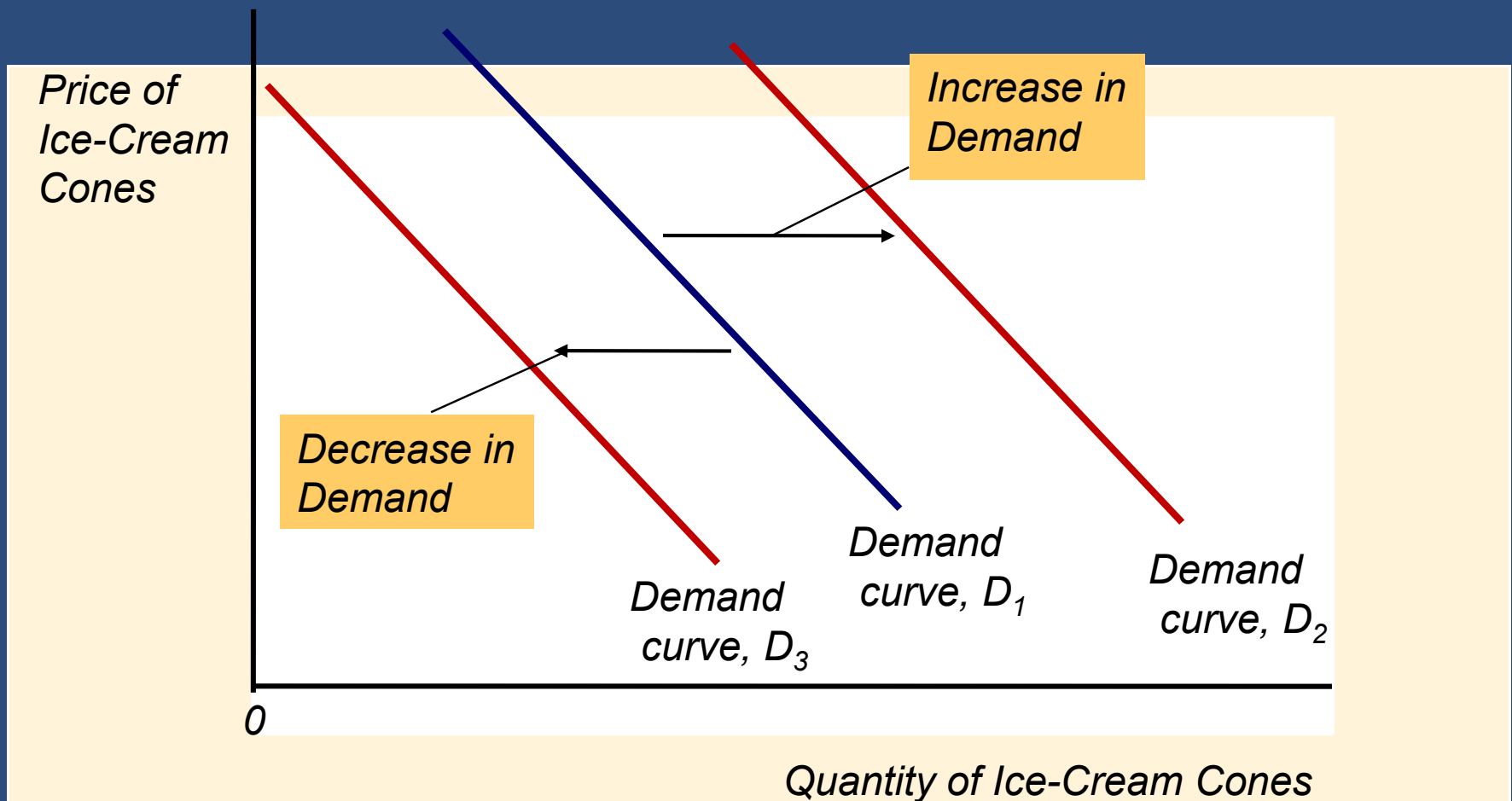
- Variables that can shift the demand curve
 - Income
 - Prices of related goods
 - Tastes
 - Expectations
 - Number of buyers



Shifts in the Demand Curve

- Change in Demand
 - A shift in the demand curve, either to the left or right.
 - Caused by any change that alters the quantity demanded at every price.

Shifts in the Demand Curve

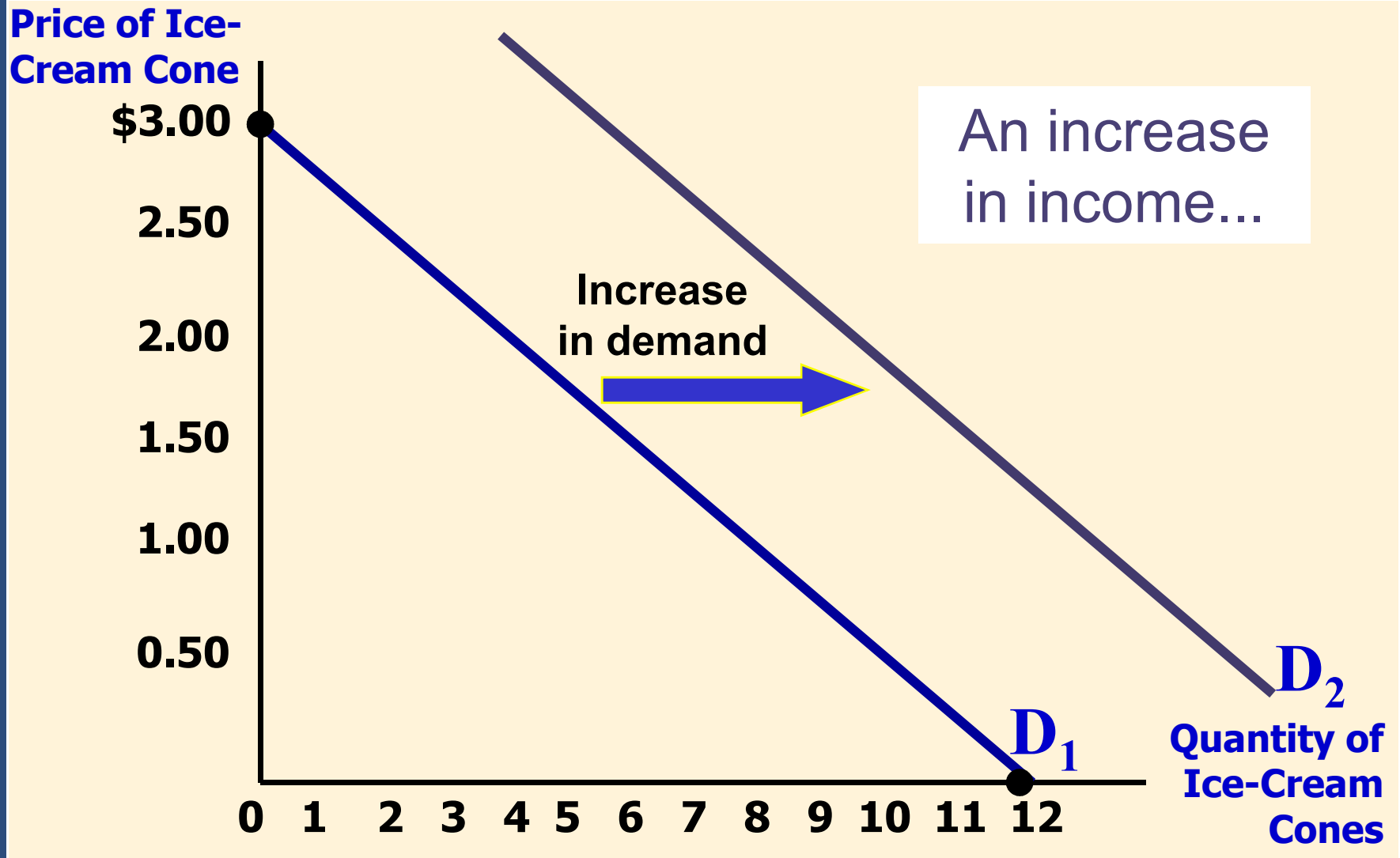


Any change that raises the quantity that buyers wish to purchase at any given price shifts the demand curve to the right. Any change that lowers the quantity that buyers wish to purchase at any given price shifts the demand curve to the left.

Shifts in the Demand Curve

- Consumer Income
 - As income increases the demand for a *normal good* will increase.
 - As income increases the demand for an *inferior good* will decrease.

Consumer Income Normal Good



Consumer Income Inferior Good

Price of Ice-
Cream Cone

\$3.00

2.50

2.00

1.50

1.00

0.50

An increase
in income...

Decrease
in demand

D_2

D_1

Quantity of
Ice-Cream
Cones

0 1 2 3 4 5 6 7 8 9 10 11 12

Shifts in the Demand Curve

- Prices of Related Goods
 - When a fall in the price of one good reduces the demand for another good, the two goods are called *substitutes*.
 - When a fall in the price of one good increases the demand for another good, the two goods are called *complements*.

Demand

- Tastes
 - Change in tastes – changes the demand
- Expectations about the future
 - Expect an increase in income
 - Increase in current demand
 - Expect higher prices
 - Increase in current demand
- Number of buyers – increase
 - Market demand - increases

Table 1 Variables That Influence Buyers

Variable	A Change in This Variable . . .
Price	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

This table lists the variables that affect how much consumers choose to buy of any good. Notice the special role that the price of the good plays:

A change in the good's price represents a movement along the demand curve, whereas a change in one of the other variables shifts the demand curve.

Two ways to reduce the quantity of smoking demanded

1. Shift the demand curve for cigarettes and other tobacco products
 - Public service announcements
 - Mandatory health warnings on cigarette packages
 - Prohibition of cigarette advertising on television
- If successful
 - Shift demand curve to the left

Two ways to reduce the quantity of smoking demanded

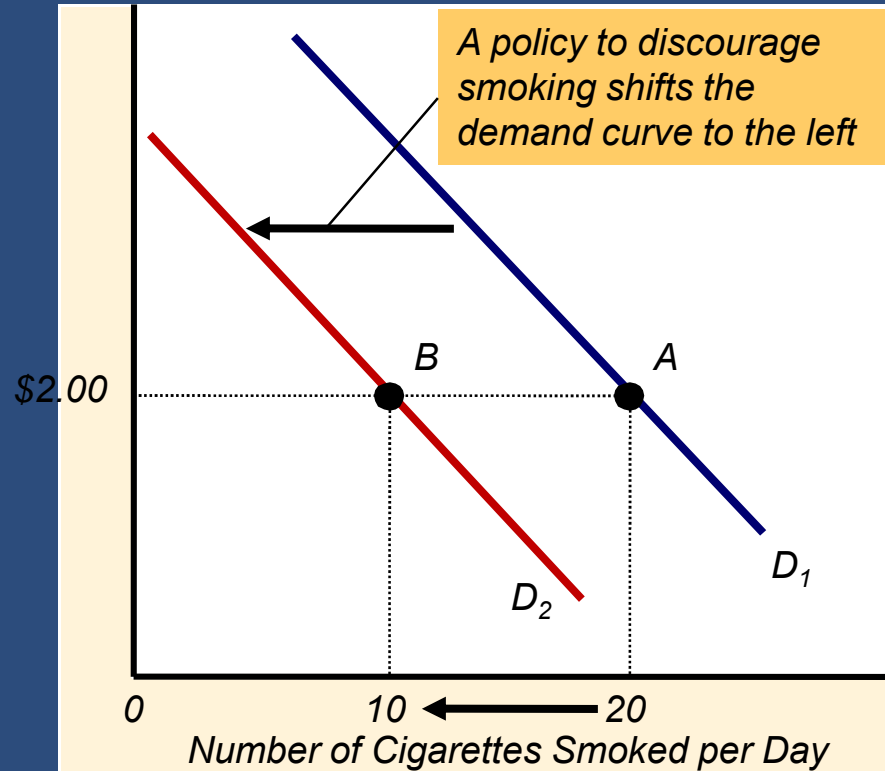
2. Try to raise the price of cigarettes

- Tax the manufacturer
 - Higher price
- Movement along demand curve
 - 10% $\uparrow$ in price $\longrightarrow$ 4% $\downarrow$ in smoking
 - Teenagers: 10% $\uparrow$ in price $\longrightarrow$ 12% $\downarrow$ in smoking

Shifts in the Demand Curve versus Movements along the Demand Curve

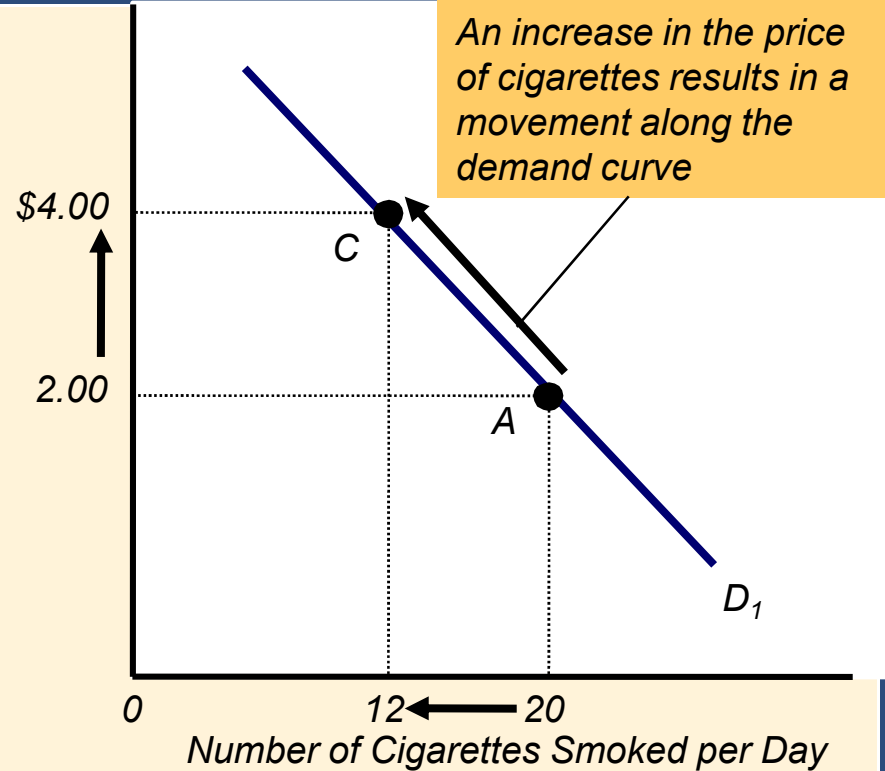
(a) A Shift in the Demand Curve

Price of Cigarettes, per Pack



(b) A Movement along the Demand Curve

Price of Cigarettes, per Pack



If warnings on cigarette packages convince smokers to smoke less, the demand curve for cigarettes shifts to the left. In panel (a), the demand curve shifts from D_1 to D_2 . At a price of \$2.00 per pack, the quantity demanded falls from 20 to 10 cigarettes per day, as reflected by the shift from point A to point B. By contrast, if a tax raises the price of cigarettes, the demand curve does not shift. Instead, we observe a movement to a different point on the demand curve. In panel (b), when the price rises from \$2.00 to \$4.00, the quantity demanded falls from 20 to 12 cigarettes per day, as reflected by the movement from point A to point C.

SUPPLY

- *Quantity supplied* is the amount of a good that sellers are willing and able to sell.
- Law of Supply
 - The *law of supply* states that, other things equal, the quantity supplied of a good rises when the price of the good rises.

The Supply Curve: The Relationship between Price and Quantity Supplied

- Supply Schedule
 - The *supply schedule* is a table that shows the relationship between the price of the good and the quantity supplied.

Ben's Supply Schedule

Price of Ice-Cream Cone	Quantity of Cones Demanded
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\$0.00

0

0.50

0

1.00

1

1.50

2

2.00

3

2.50

4

3.00

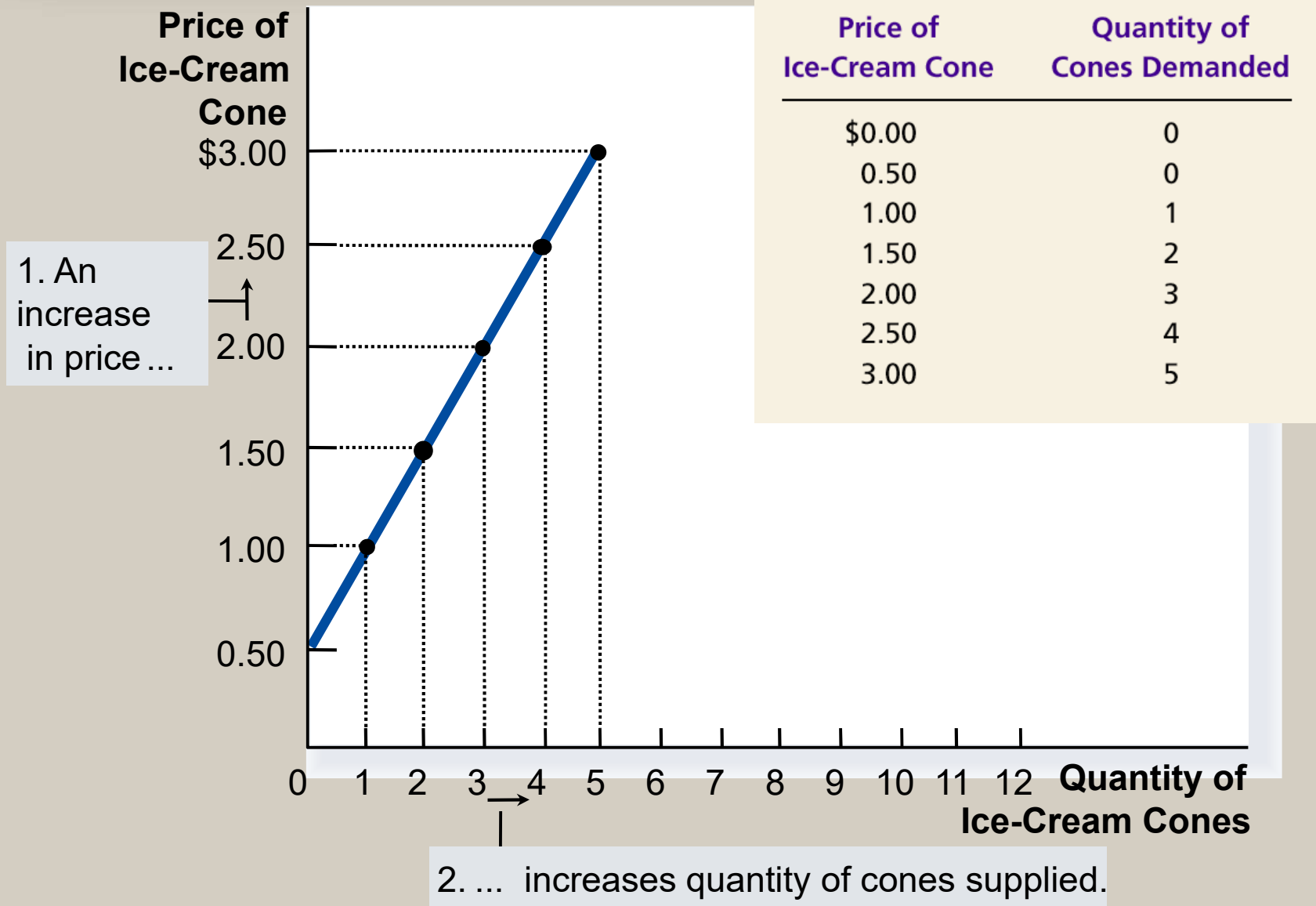
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The Supply Curve: The Relationship between Price and Quantity Supplied

- Supply Curve
 - The *supply curve* is the graph of the relationship between the price of a good and the quantity supplied.

Figure 5 Ben's Supply Schedule and Supply Curve



Market Supply versus Individual Supply

- Market supply refers to the sum of all individual supplies for all sellers of a particular good or service.
- Graphically, individual supply curves are summed horizontally to obtain the market supply curve.

Market Supply as the Sum of Individual Supplies

Ben's supply

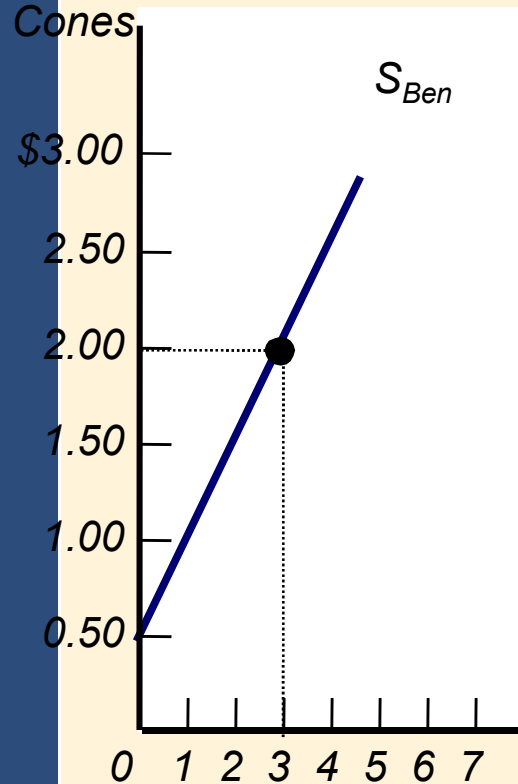
+

Jerry's supply

=

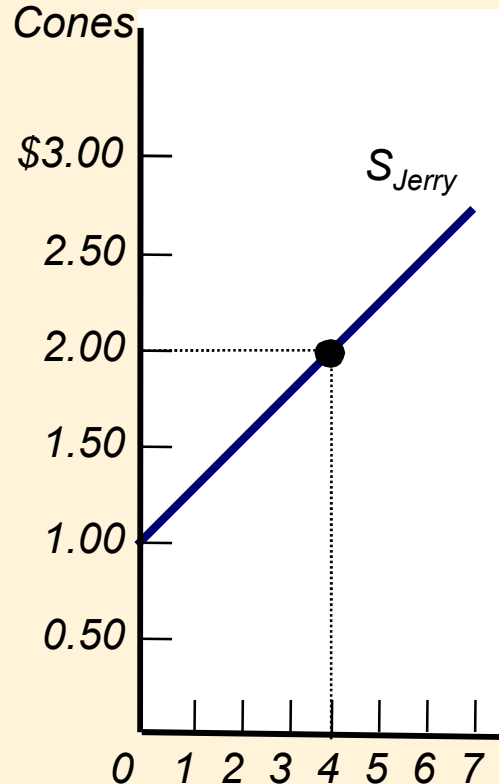
Market supply

Price of
Ice-Cream
Cones



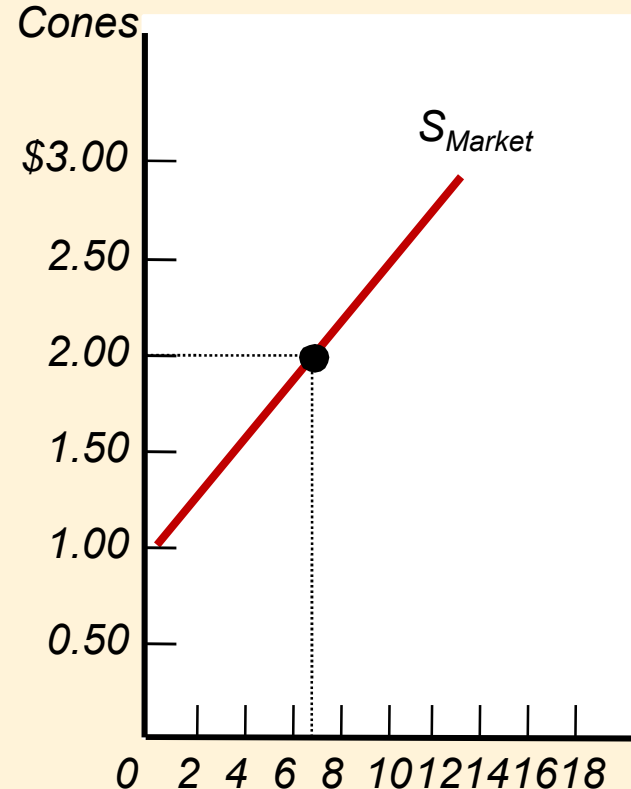
Quantity of
Ice-Cream Cones

Price of
Ice-Cream
Cones



Quantity of
Ice-Cream Cones

Price of
Ice-Cream
Cones



Quantity of
Ice-Cream Cones

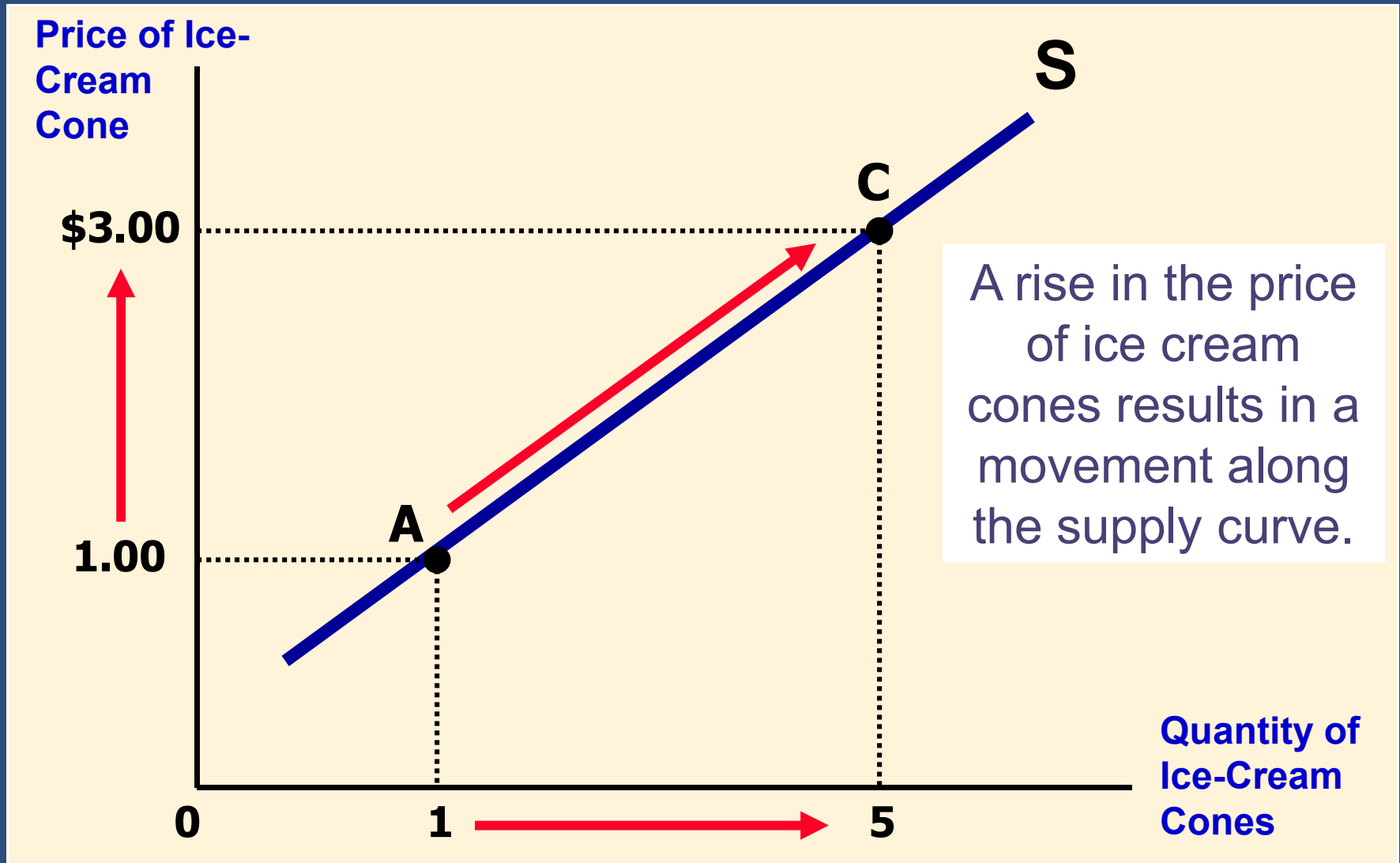
Table 2.4: Derivation of the market supply of good X

Price per unit	Quantity supplied by seller 1	Quantity supplied by seller 2	Quantity supplied by seller 3	Market supply per week
5	11	15	8	34
4	10.5	13	7	30.5
3	8	11.5	5.5	25
2	6	8.5	4	18.5
1	4	6	2	12

Shifts in the Supply Curve

- Change in Quantity Supplied
 - Movement along the supply curve.
 - Caused by a change in anything that alters the quantity supplied at each price.

Change in Quantity Supplied



Shifts in the Supply Curve

- Change in Supply
 - A shift in the supply curve, either to the left or right.
 - Caused by a change in a determinant other than price.

Shifts in the Supply Curve

- Input prices
- Technology
- Expectations
- Number of sellers



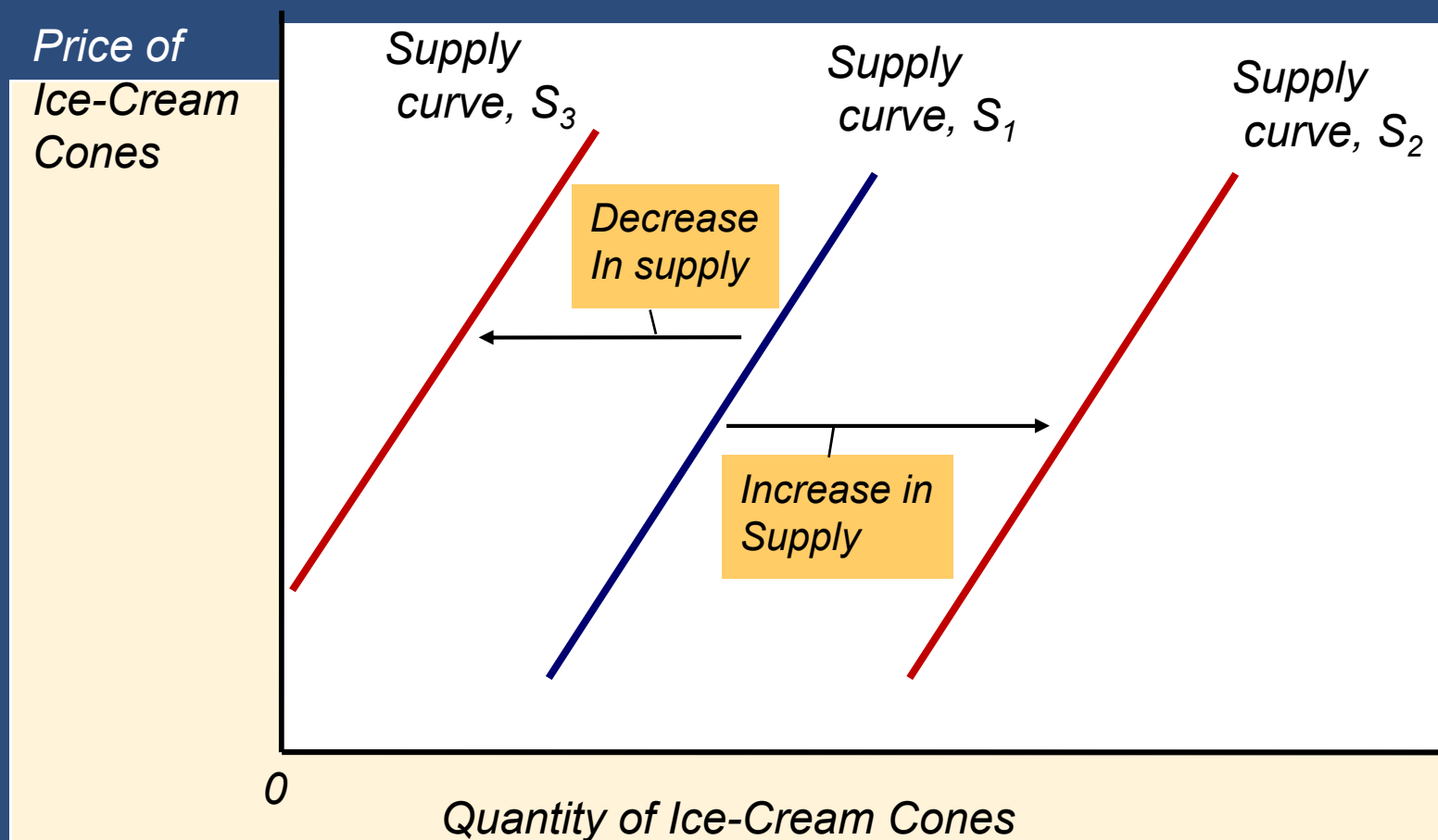
Supply

- Input Prices (costs)
 - Supply – negatively related to prices of inputs
 - Higher input prices – decrease in supply
- Technology
 - Advance in technology – increase in supply

Supply

- Expectations about future
 - Affect current supply
 - Expected higher prices
 - Decrease in current supply
- Number of sellers – increase
 - Market supply - increase

Shifts in the Supply Curve



Any change that raises the quantity that sellers wish to produce at any given price shifts the supply curve to the right. Any change that lowers the quantity that sellers wish to produce at any given price shifts the supply curve to the left.

Variables That Influence Sellers

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

This table lists the variables that affect how much producers choose to sell of any good. Notice the special role that the price of the good plays: A change in the good's price represents a movement along the supply curve, whereas a change in one of the other variables shifts the supply curve.

SUPPLY AND DEMAND TOGETHER

- *Equilibrium* refers to a situation in which the price has reached the level where quantity supplied equals quantity demanded.

SUPPLY AND DEMAND TOGETHER

- *Equilibrium Price*

- The price that balances quantity supplied and quantity demanded.
- On a graph, it is the price at which the supply and demand curves intersect.

- *Equilibrium Quantity*

- The quantity supplied and the quantity demanded at the equilibrium price.
- On a graph it is the quantity at which the supply and demand curves intersect.

SUPPLY AND DEMAND TOGETHER

Demand Schedule

Price of Ice-Cream Cone	Market
\$0.00	19
0.50	16
1.00	13
1.50	10
2.00	7
2.50	4
3.00	1

Supply Schedule

Price of Ice-Cream Cone	Market
\$0.00	0
0.50	0
1.00	1
1.50	4
2.00	7
2.50	10
3.00	13

At \$2.00, the quantity demanded is equal to the quantity supplied!

Figure 8 The Equilibrium of Supply and Demand

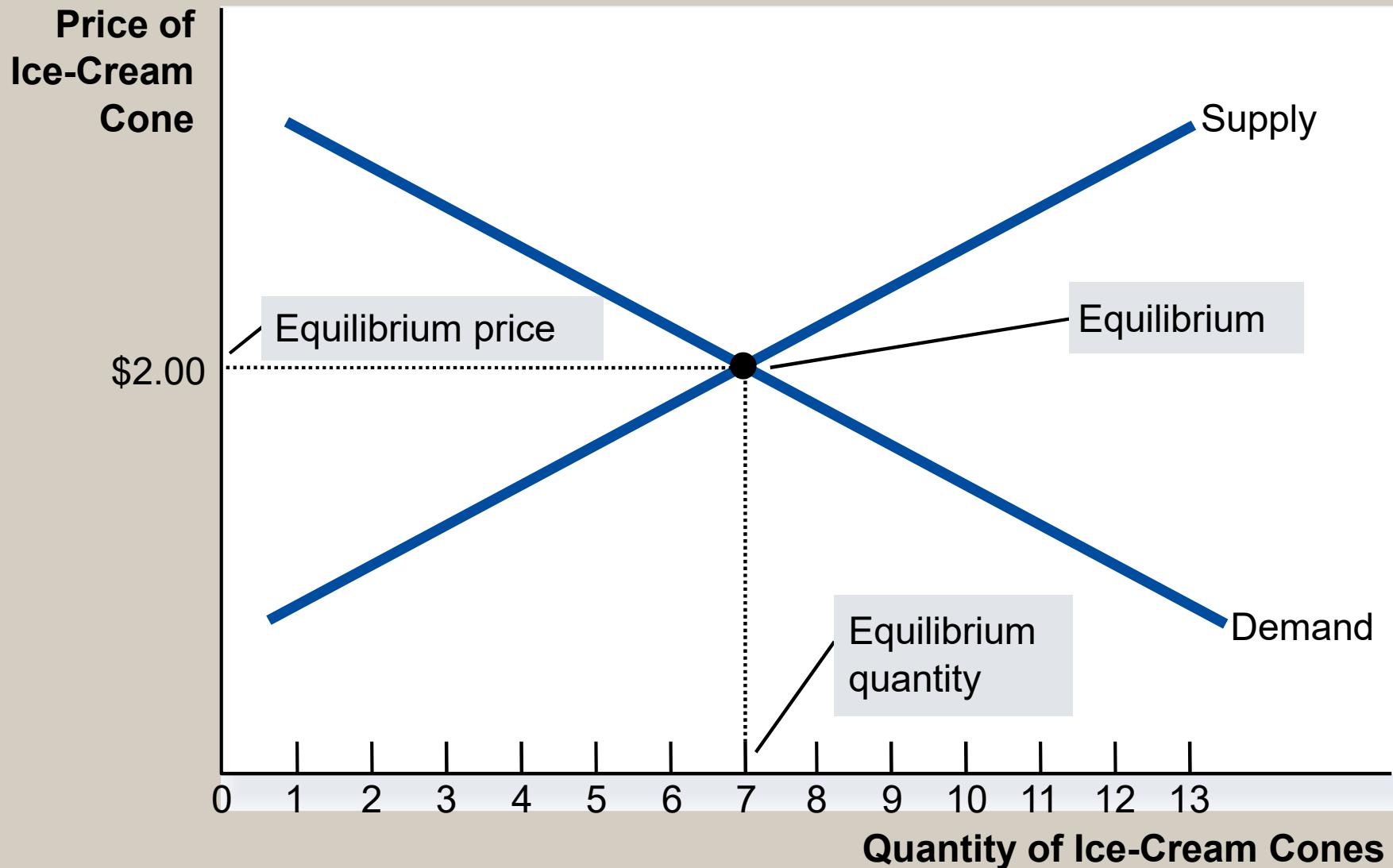
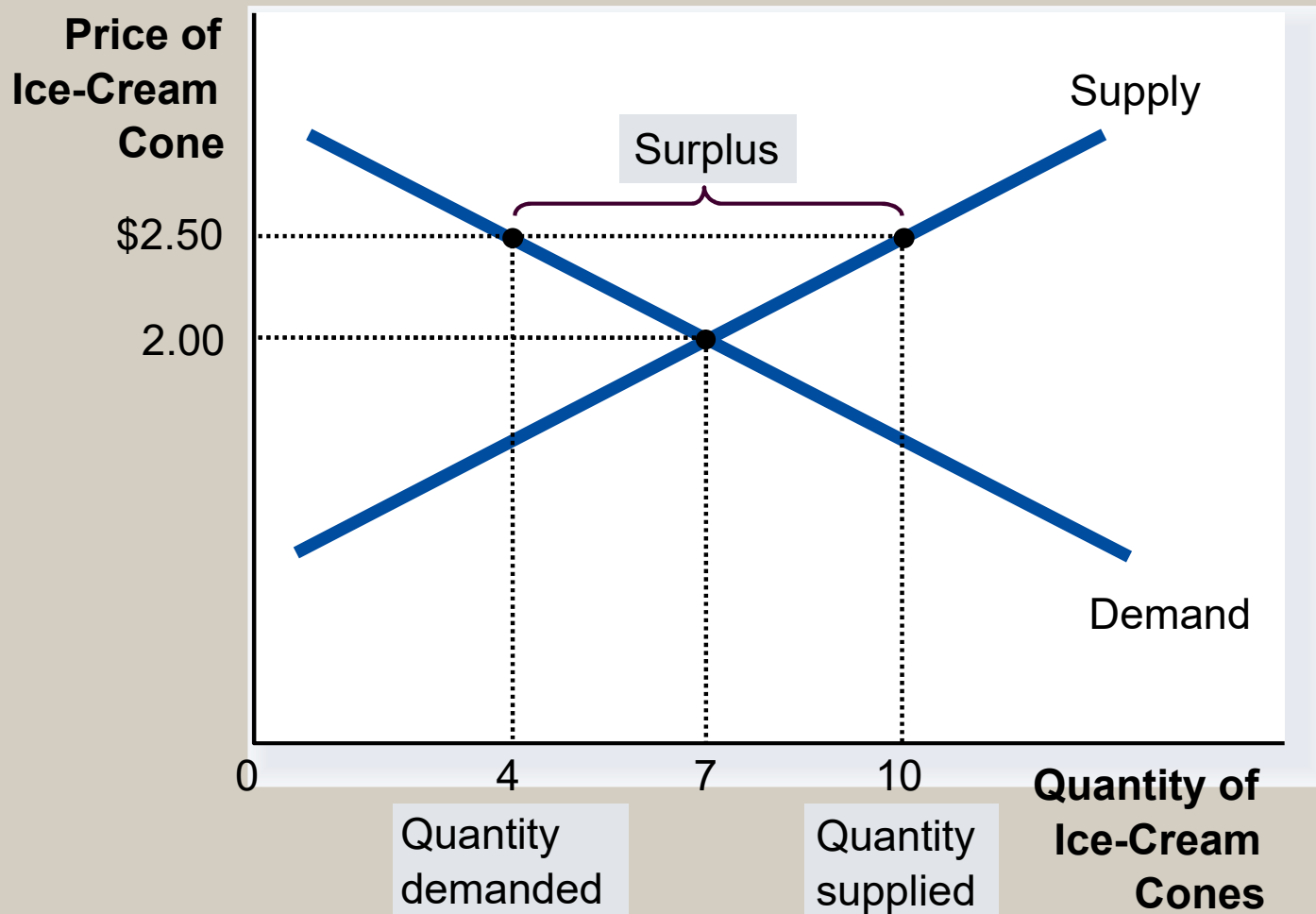


Figure 9 Markets Not in Equilibrium

(a) Excess Supply



Equilibrium

- *Surplus*

- When $\text{price} > \text{equilibrium price}$, then $\text{quantity supplied} > \text{quantity demanded}$.
 - There is excess supply or a surplus.
 - Suppliers will lower the price to increase sales, thereby moving toward equilibrium.

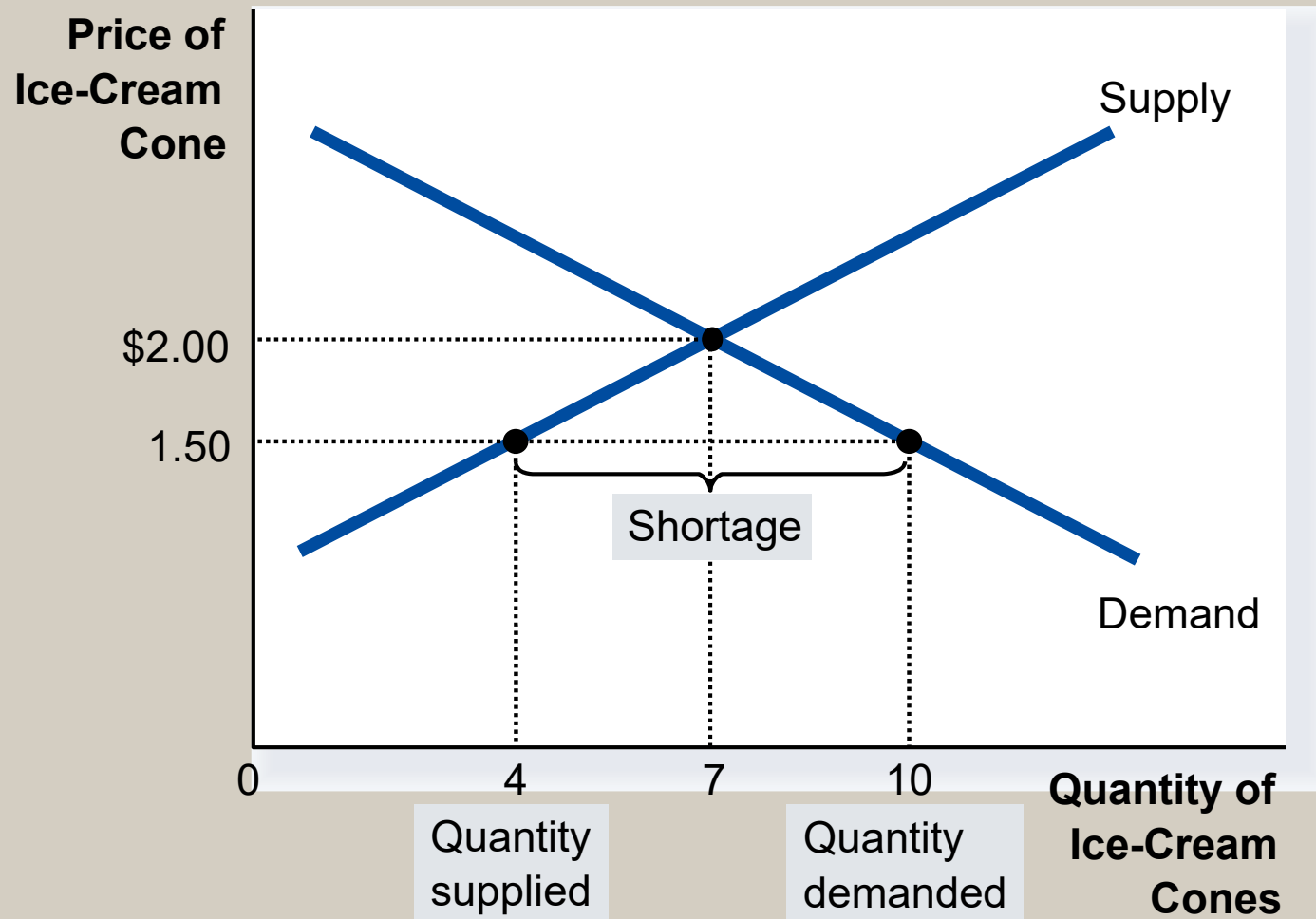
Equilibrium

- *Shortage*

- When $\text{price} < \text{equilibrium price}$, then $\text{quantity demanded} > \text{the quantity supplied}$.
 - There is excess demand or a shortage.
 - Suppliers will raise the price due to too many buyers chasing too few goods, thereby moving toward equilibrium.

Figure 9 Markets Not in Equilibrium

(b) Excess Demand



Equilibrium

- *Law of supply and demand*
 - The claim that the price of any good adjusts to bring the quantity supplied and the quantity demanded for that good into balance.
 - In most markets
 - Surpluses and shortages are temporary

Three Steps to Analyzing Changes in Equilibrium

- Decide whether the event shifts the supply or demand curve (or both).
- Decide whether the curve(s) shift(s) to the left or to the right.
- Use the supply-and-demand diagram to see how the shift affects equilibrium price and quantity.

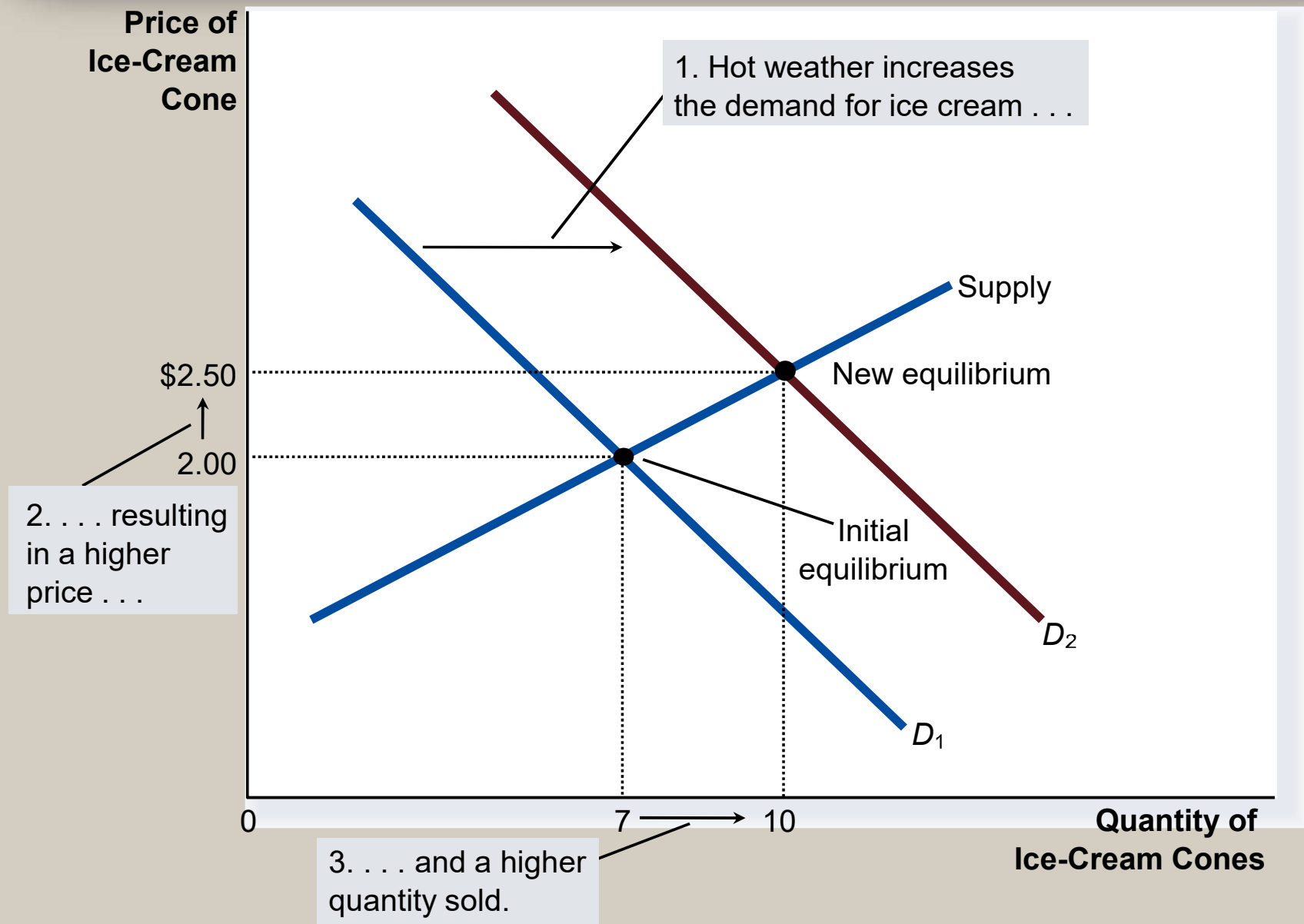
Three Steps to Analyzing Changes in Equilibrium

- Shifts in Curves versus Movements along Curves
 - A shift in the supply curve is called a change in supply.
 - A movement along a fixed supply curve is called a change in quantity supplied.
 - A shift in the demand curve is called a change in demand.
 - A movement along a fixed demand curve is called a change in quantity demanded.

Supply and Demand Together

- A change in market equilibrium due to a shift in demand
 - One summer - very hot weather
 - Effect on the market for ice cream?
 1. Hot weather – shifts the demand curve (tastes)
 2. Demand curve shifts to the right
 3. Higher equilibrium price; higher equilibrium quantity

Figure 10 How an Increase in Demand Affects the Equilibrium



Supply and Demand Together

- A change in market equilibrium due to a shift in supply
 - One summer - a hurricane destroys part of the sugarcane crop: higher price of sugar
 - Effect on the market for ice cream?
 1. Change in price of sugar - supply curve
 2. Supply curve - shifts to the left
 3. Higher equilibrium price; lower equilibrium quantity

How a Decrease in Supply Affects the Equilibrium

Price of
Ice-Cream
Cones

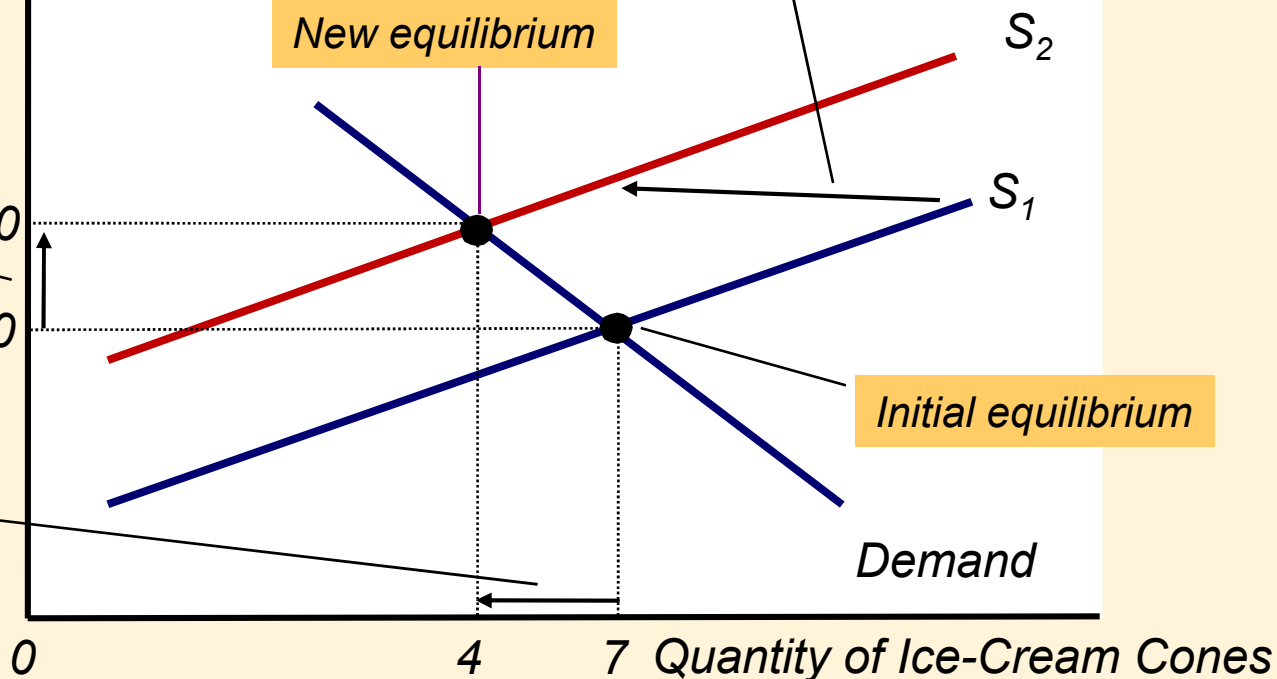
2. ...resulting in
a higher price ...

3. ...and a smaller
quantity sold.

1. An increase in the price of sugar reduces
the supply of ice cream ...

New equilibrium

Initial equilibrium



An event that reduces quantity supplied at any given price shifts the supply curve to the left. The equilibrium price rises, and the equilibrium quantity falls. Here an increase in the price of sugar (an input) causes sellers to supply less ice cream. The supply curve shifts from S_1 to S_2 , which causes the equilibrium price of ice cream to rise from \$2.00 to \$2.50 and the equilibrium quantity to fall from 7 to 4 cones.

How Prices Allocate Resources

- Prices
 - Signals that guide the allocation of resources
 - Mechanism for rationing scarce resources
 - Determine who produces each good and how much is produced

Summary

- Economists use the model of supply and demand to analyze competitive markets.
- In a competitive market, there are many buyers and sellers, each of whom has little or no influence on the market price.

Summary

- The demand curve shows how the quantity of a good depends upon the price.
 - According to the law of demand, as the price of a good falls, the quantity demanded rises. Therefore, the demand curve slopes downward.
 - In addition to price, other determinants of how much consumers want to buy include income, the prices of complements and substitutes, tastes, expectations, and the number of buyers.
 - If one of these factors changes, the demand curve shifts.

Summary

- The supply curve shows how the quantity of a good supplied depends upon the price.
 - According to the law of supply, as the price of a good rises, the quantity supplied rises. Therefore, the supply curve slopes upward.
 - In addition to price, other determinants of how much producers want to sell include input prices, technology, expectations, and the number of sellers.
 - If one of these factors changes, the supply curve shifts.

Summary

- Market equilibrium is determined by the intersection of the supply and demand curves.
- At the equilibrium price, the quantity demanded equals the quantity supplied.
- The behavior of buyers and sellers naturally drives markets toward their equilibrium.

Summary

- To analyze how any event influences a market, we use the supply-and-demand diagram to examine how the event affects the equilibrium price and quantity.
- In market economies, prices are the signals that guide economic decisions and thereby allocate resources.



2.2

Elasticity and Its Applications

Elasticity . . .

- ... allows us to analyze supply and demand with greater precision.
- ... is a measure of how much buyers and sellers respond to changes in market conditions

Elasticity of Demand

- **Elasticity measures the extent to which demand will change.**
- Measurement of the percentage change in one variable that results from a 1% change in another variable.

Types of elasticity of demand

Three basic types used:

- **Price elasticity of demand**
- **Income elasticity of demand**
- **Cross elasticity**

THE ELASTICITY OF DEMAND

- *Price elasticity of demand* is a measure of how much the quantity demanded of a good responds to a change in the price of that good.
- Price elasticity of demand is the percentage change in quantity demanded given a percent change in the price.

Computing the Price Elasticity of Demand

- The price elasticity of demand is computed as the percentage change in the quantity demanded divided by the percentage change in price.

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

Computing the Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- Example: If the price of an ice cream cone increases from \$2.00 to \$2.20 and the amount you buy falls from 10 to 8 cones, then your elasticity of demand would be calculated as:

$$\frac{\frac{(10 - 8)}{10} \times 100}{\frac{(2.20 - 2.00)}{2.00} \times 100} = \frac{20\%}{10\%} = 2$$

The Midpoint Method: A Better Way to Calculate Percentage Changes and Elasticities

- The midpoint formula is preferable when calculating the price elasticity of demand because it gives the same answer regardless of the direction of the change.

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1) / [(Q_2 + Q_1) / 2]}{(P_2 - P_1) / [(P_2 + P_1) / 2]}$$

The Midpoint Method: A Better Way to Calculate Percentage Changes and Elasticities

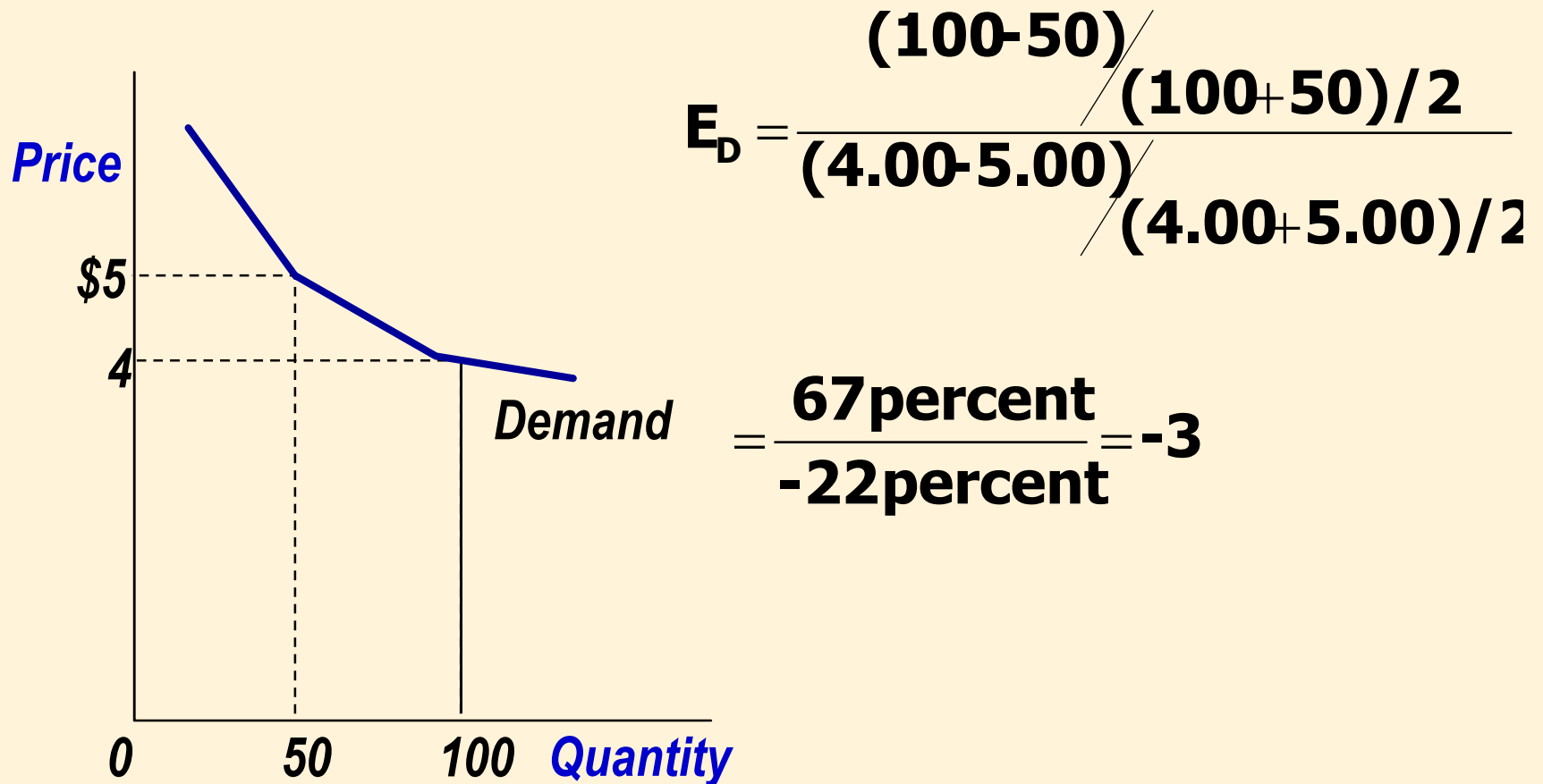
- Example: If the price of an ice cream cone increases from \$2.00 to \$2.20 and the amount you buy falls from 10 to 8 cones, then your elasticity of demand, using the midpoint formula, would be calculated as:

$$\frac{\frac{(10 - 8)}{(10 + 8) / 2}}{\frac{(2.20 - 2.00)}{(2.00 + 2.20) / 2}} = \frac{22\%}{9.5\%} = 2.32$$

The Variety of Demand Curves

- Inelastic Demand
 - Quantity demanded does not respond strongly to price changes.
 - Price elasticity of demand is less than one.
- Elastic Demand
 - Quantity demanded responds strongly to changes in price.
 - Price elasticity of demand is greater than one.

Computing the Price Elasticity of Demand



Demand is price elastic

The Variety of Demand Curves

- Perfectly Inelastic
 - Quantity demanded does not respond to price changes.
- Perfectly Elastic
 - Quantity demanded changes infinitely with any change in price.
- Unit Elastic
 - Quantity demanded changes by the same percentage as the price.

The Variety of Demand Curves

- Because the price elasticity of demand measures how much quantity demanded responds to the price, it is closely related to the slope of the demand curve.

Figure 1 The Price Elasticity of Demand

(a) Perfectly Inelastic Demand: Elasticity Equals 0

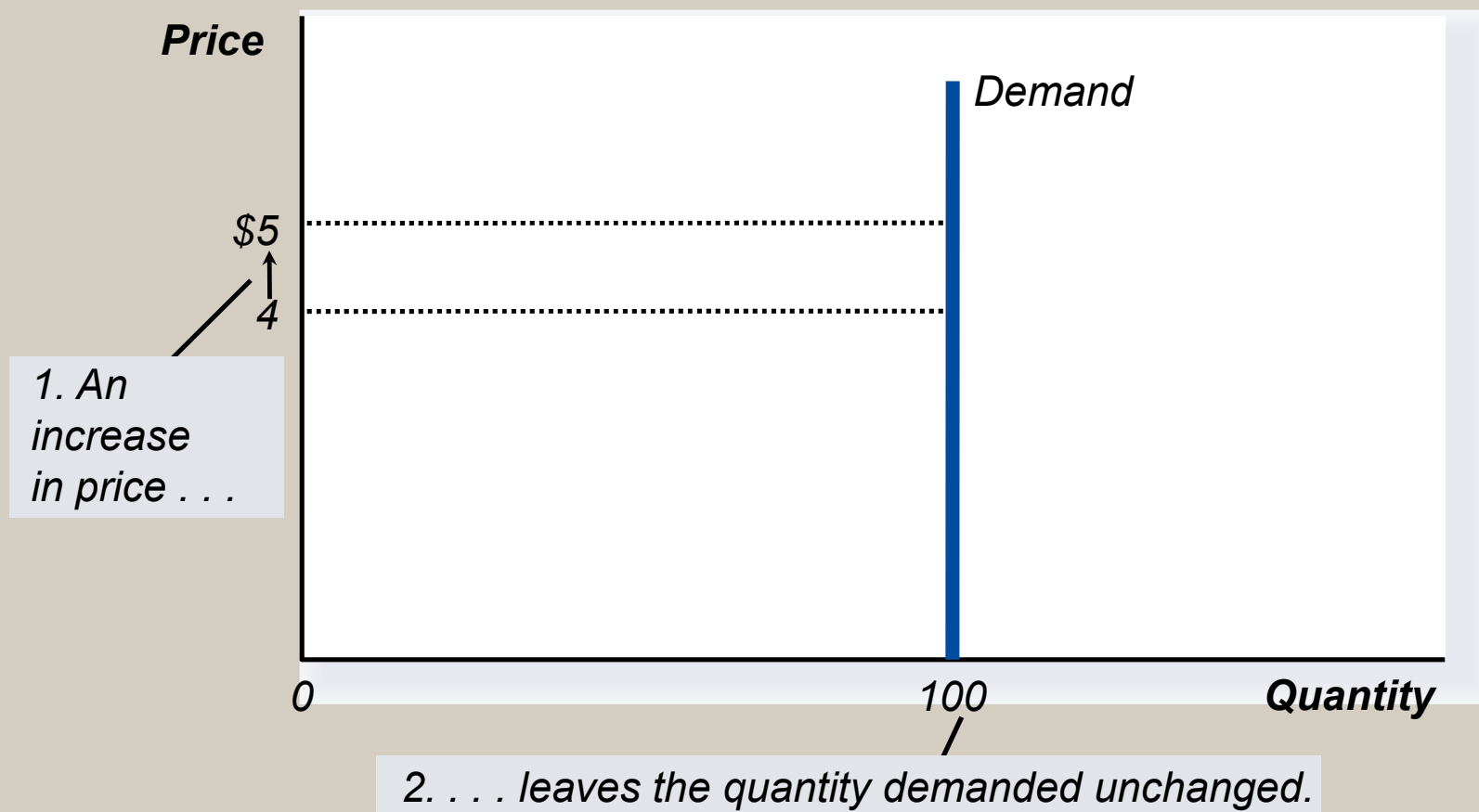


Figure 1 The Price Elasticity of Demand

(b) Inelastic Demand: Elasticity Is Less Than 1

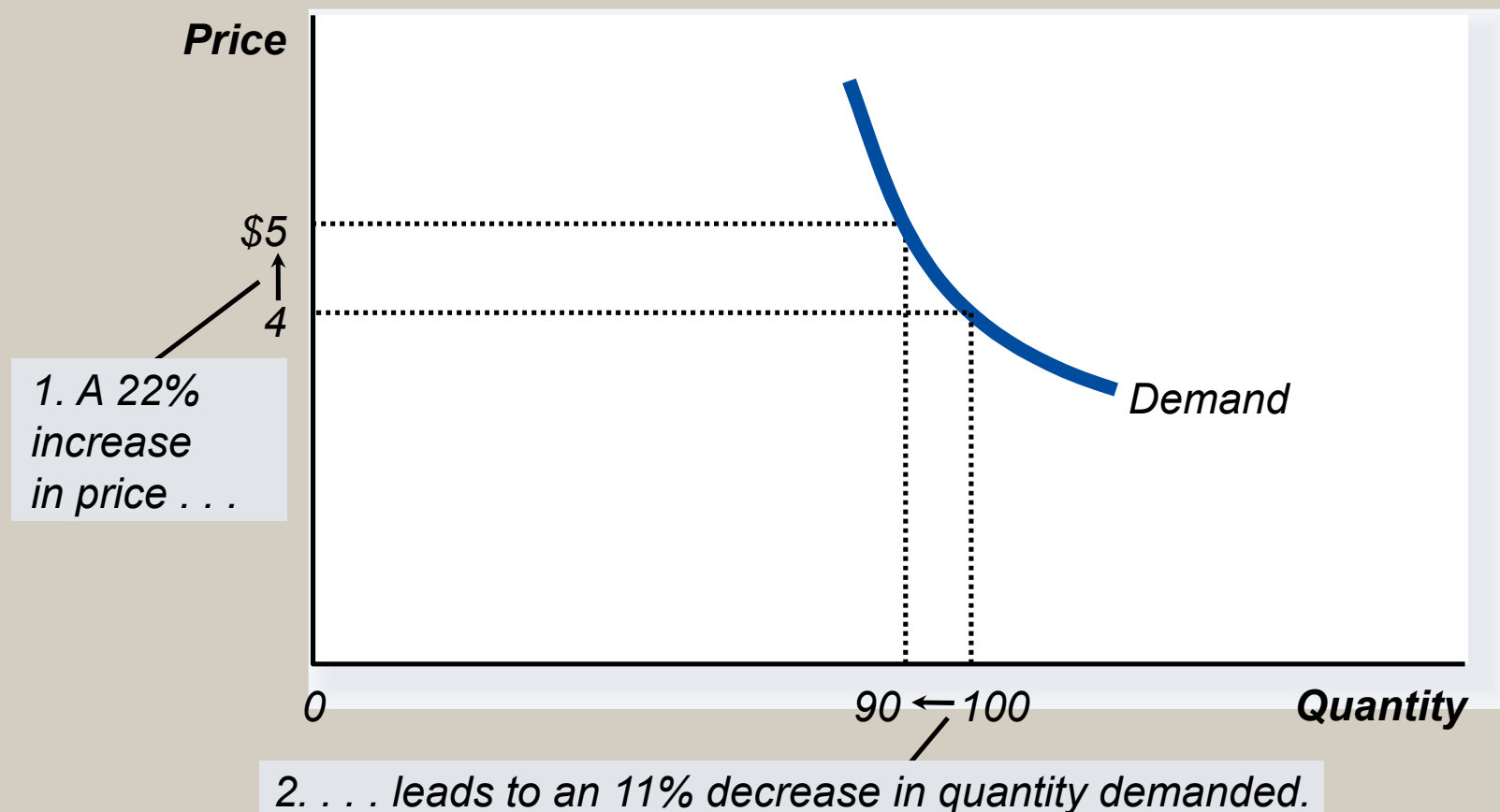


Figure 1 The Price Elasticity of Demand

(c) Unit Elastic Demand: Elasticity Equals 1

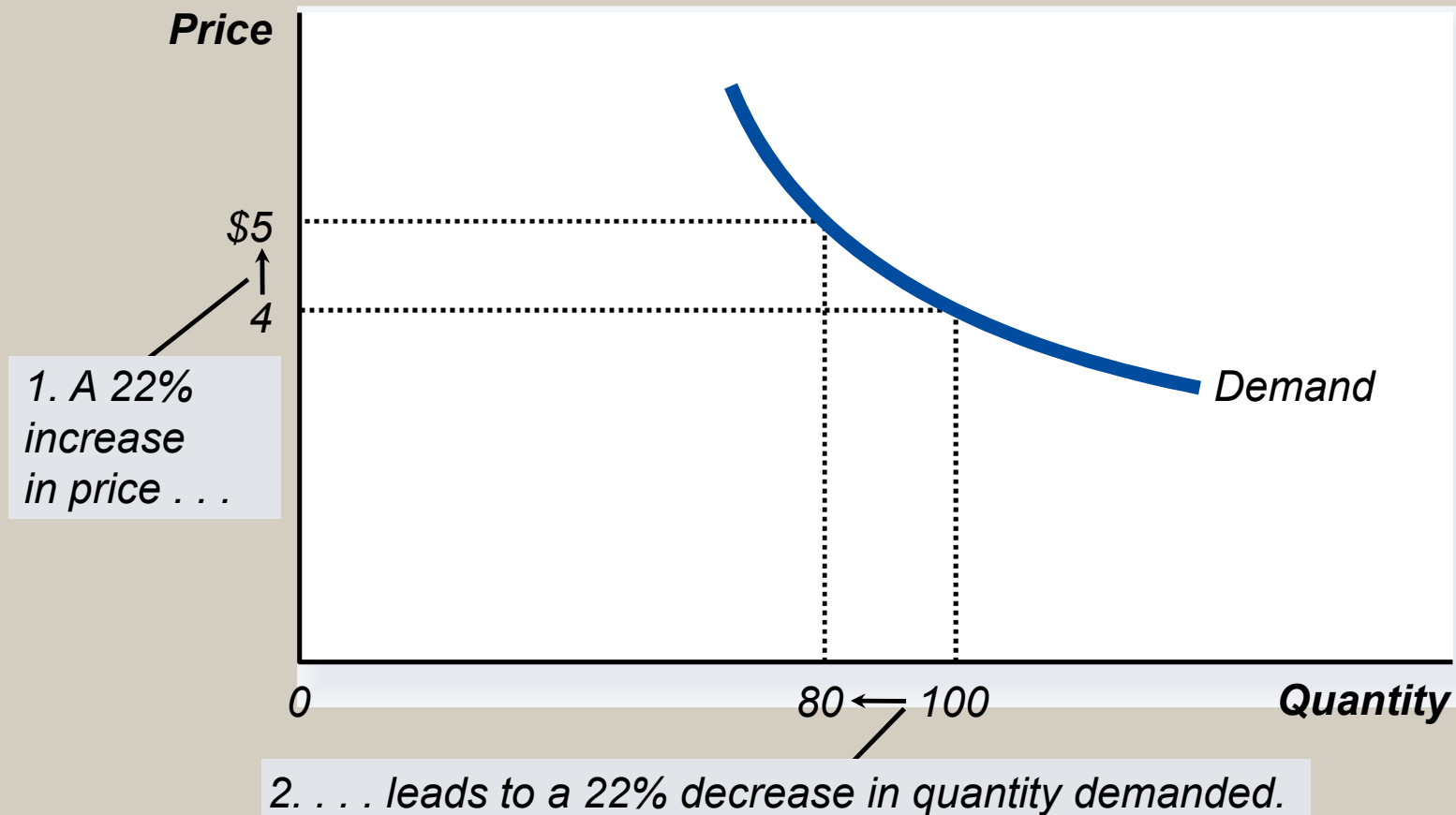


Figure 1 The Price Elasticity of Demand

(d) Elastic Demand: Elasticity Is Greater Than 1

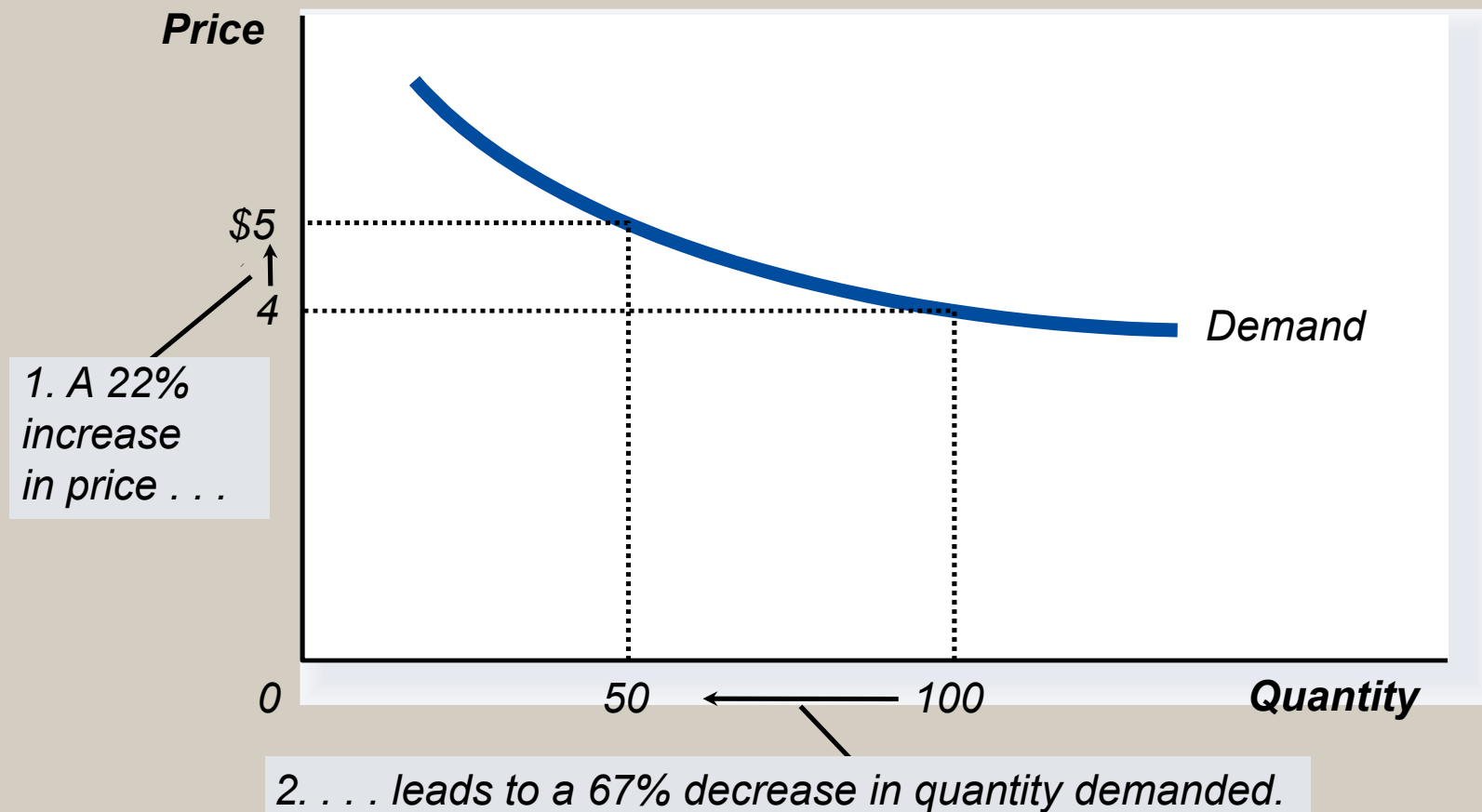
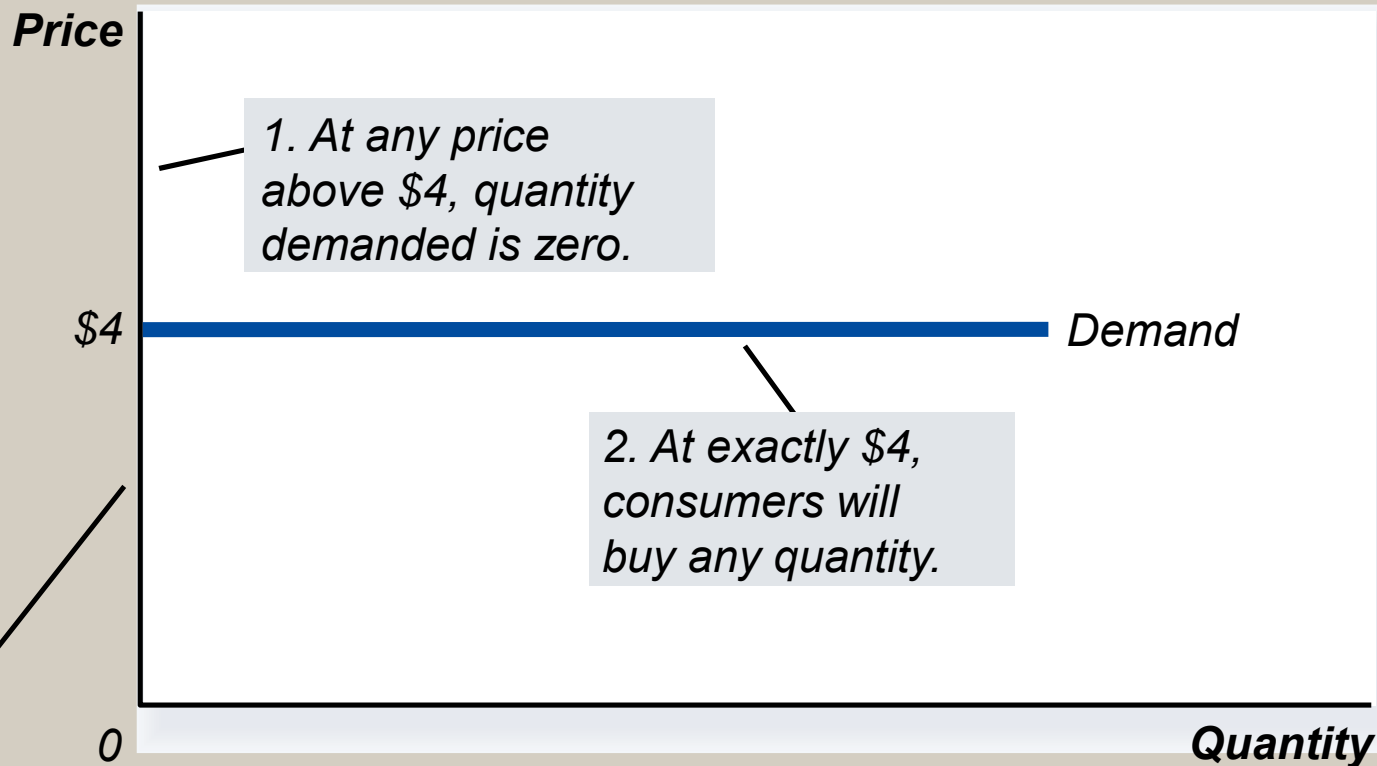


Figure 1 The Price Elasticity of Demand

(e) *Perfectly Elastic Demand: Elasticity Equals Infinity*

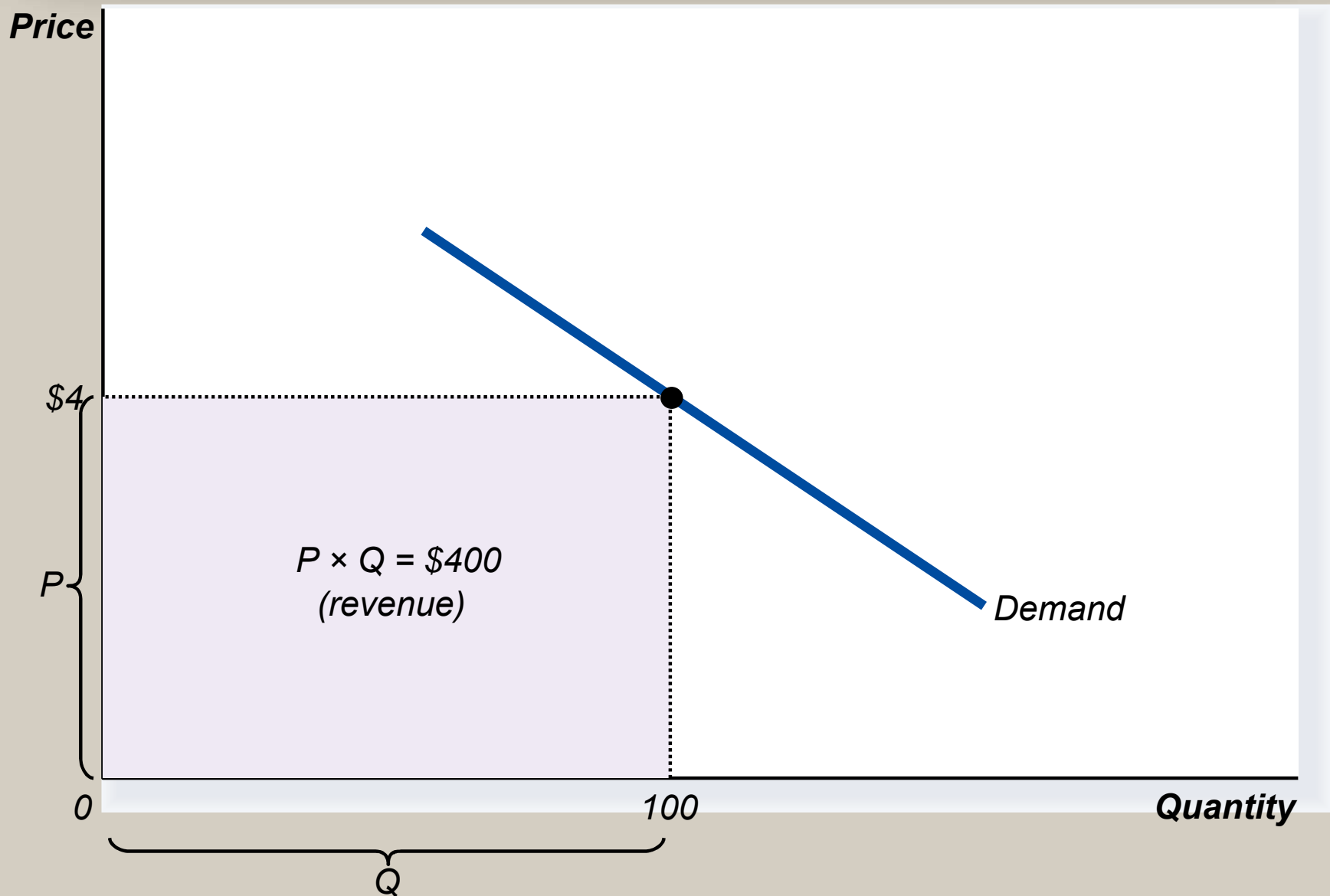


Total Revenue and the Price Elasticity of Demand

- *Total revenue* is the amount paid by buyers and received by sellers of a good.
- Computed as the price of the good times the quantity sold.

$$TR = P \times Q$$

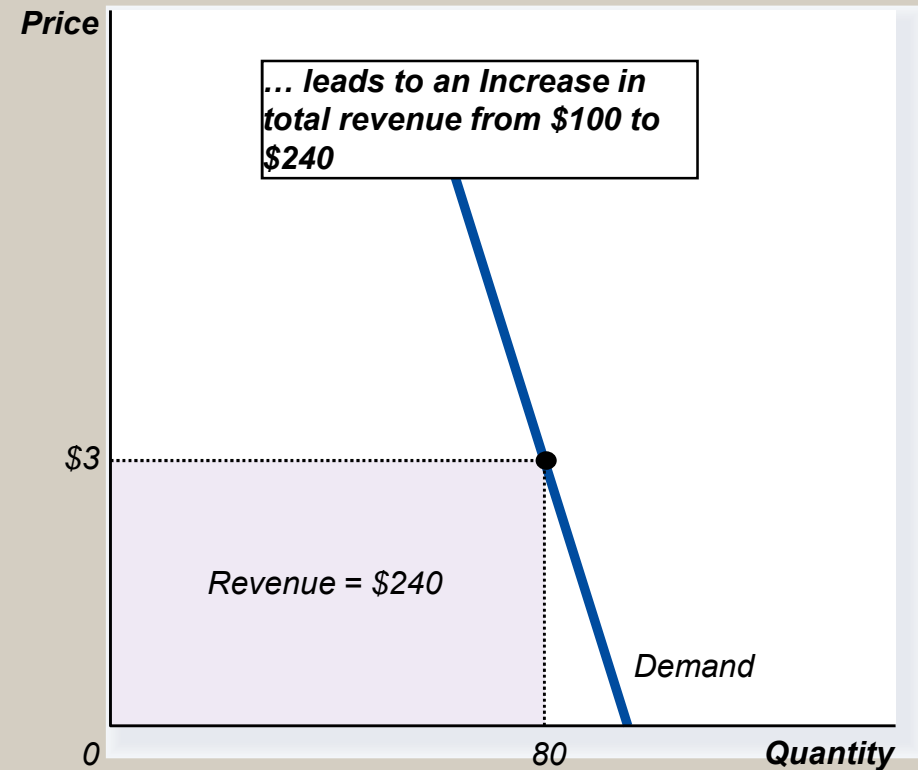
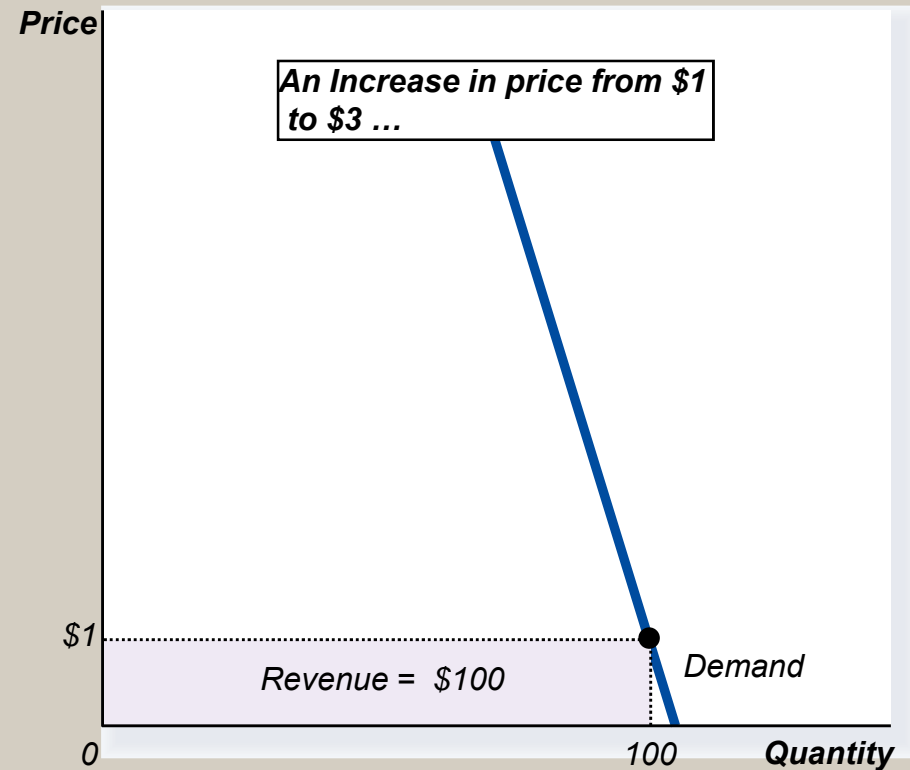
Figure 2 Total Revenue



Elasticity and Total Revenue along a Linear Demand Curve

- With an inelastic demand curve, an increase in price leads to a decrease in quantity that is proportionately smaller. Thus, *total revenue increases*.

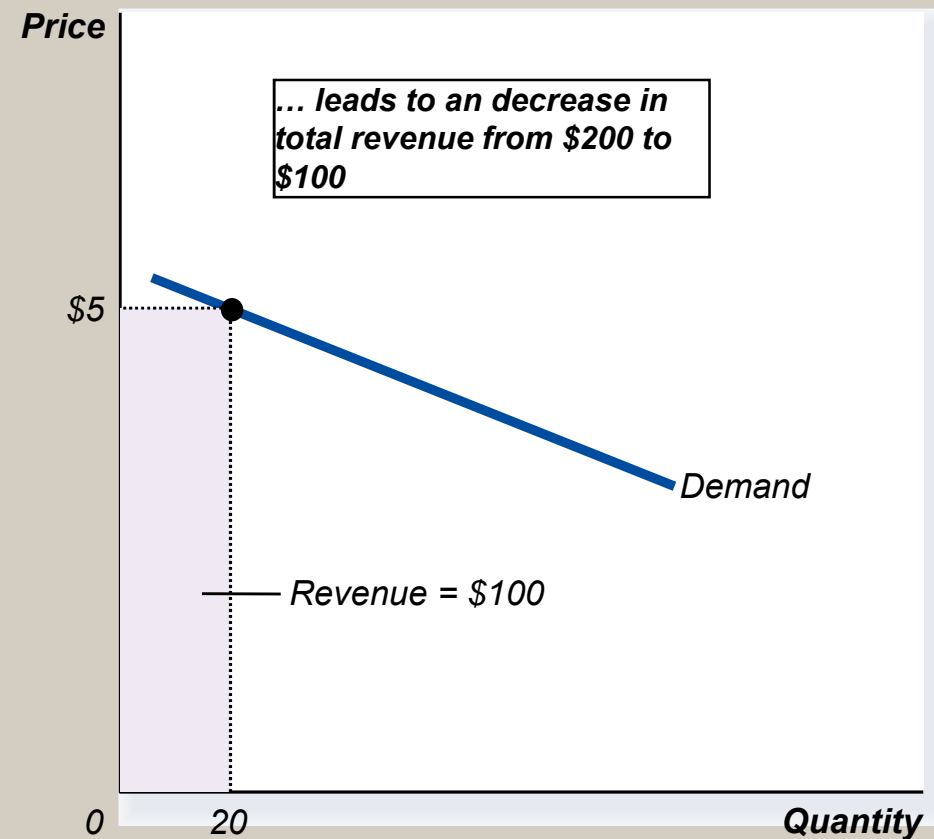
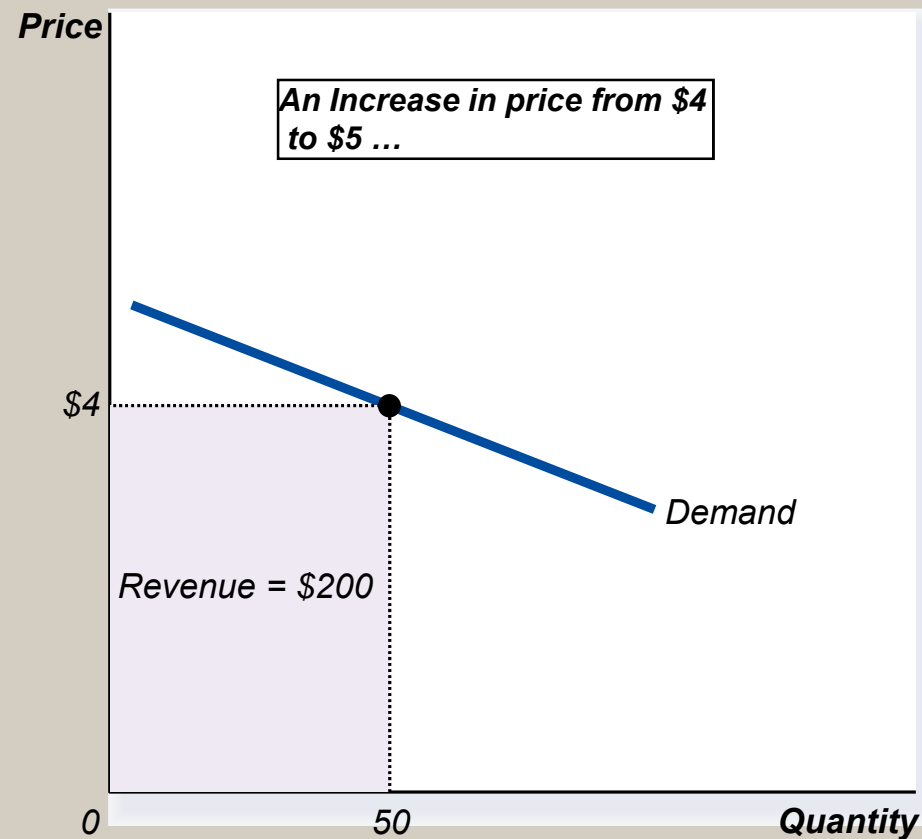
Figure 3 How Total Revenue Changes When Price Changes: Inelastic Demand



Elasticity and Total Revenue along a Linear Demand Curve

- With an elastic demand curve, an increase in the price leads to a decrease in quantity demanded that is proportionately larger. Thus, *total revenue decreases*.

Figure 4 How Total Revenue Changes When Price Changes: Elastic Demand



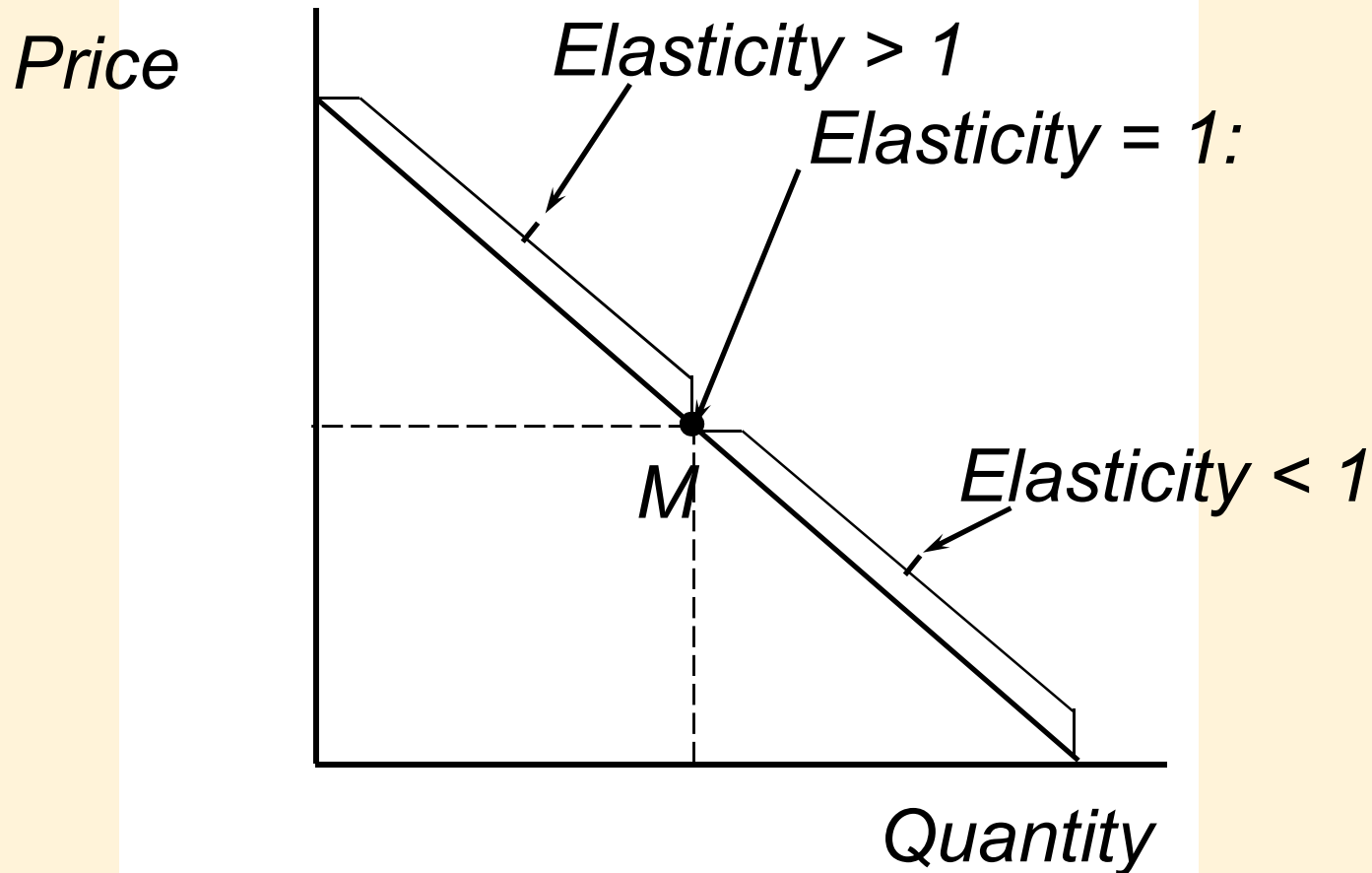
Elasticity along a Linear Demand Curve

The slope of a linear demand curve is constant, but its elasticity is not.

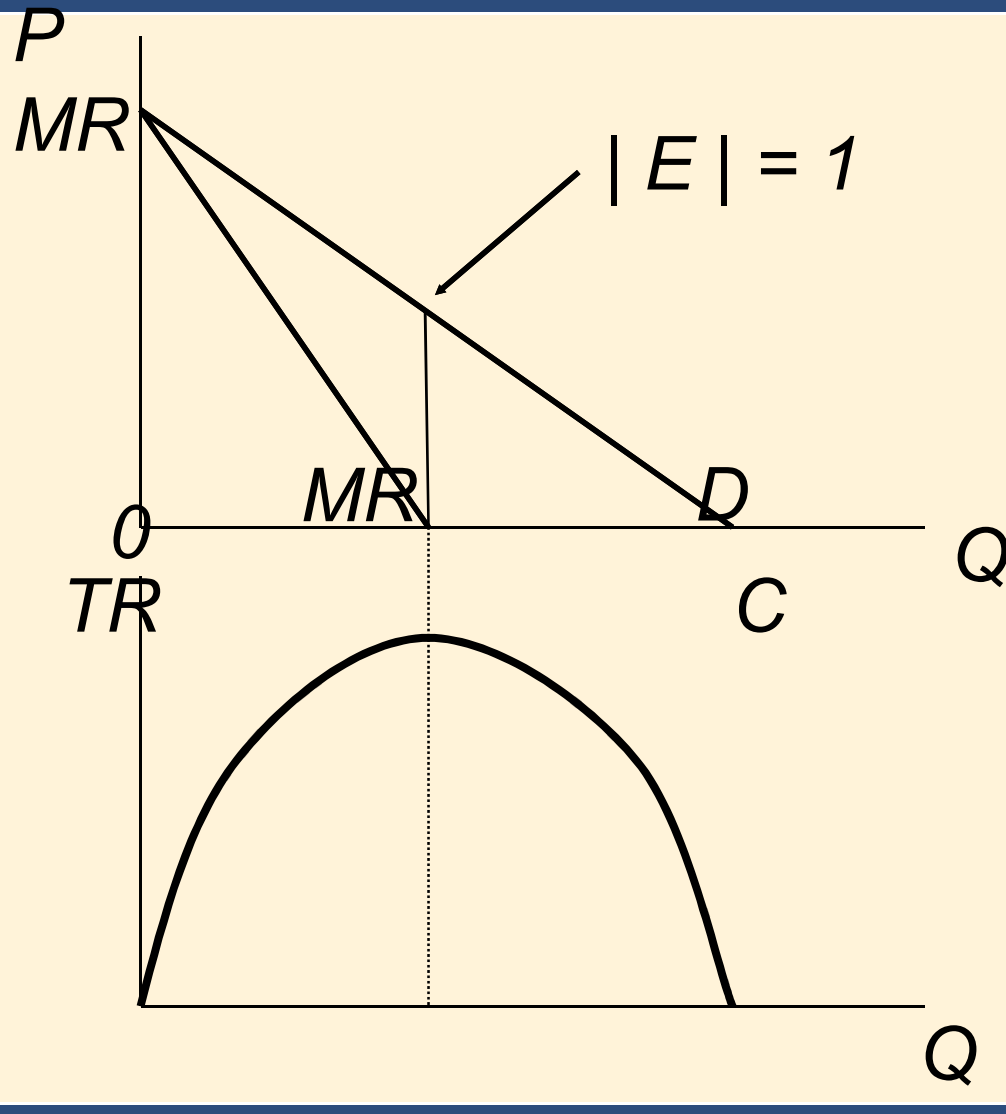
At points with a low price and high quantity, the demand curve is inelastic.

At points with a high price and low quantity, the demand curve is elastic

A linear demand curve and price elasticity of demand



Price elasticity of Demand and Total Revenue



Elasticity of a Linear Demand Curve

Price	Quantity	Total Revenue (Price × Quantity)	Percent Change in Price	Percent Change in Quantity	Elasticity	Description
\$7	0	\$ 0				
6	2	12	15	200	13.0	Elastic
5	4	20	18	67	3.7	Elastic
4	6	24	22	40	1.8	Elastic
3	8	24	29	29	1.0	Unit elastic
2	10	20	40	22	0.6	Inelastic
1	12	12	67	18	0.3	Inelastic
0	14	0	200	15	0.1	Inelastic

Price elasticity of Demand and Total Revenue

- If demand is price elastic:
 - Increasing price would **reduce** TR
($\% \Delta Q_d > \% \Delta P$)
 - Reducing price would **increase** TR
($\% \Delta Q_d > \% \Delta P$)
- If demand is price inelastic:
 - Increasing price would **increase** TR
($\% \Delta Q_d < \% \Delta P$)
 - Reducing price would **reduce** TR
($\% \Delta Q_d < \% \Delta P$)

Price elasticity, MR, AR, and TR

Total Revenue, Marginal Revenue and price elasticity

- $MR = P(1 - 1/e)$
- If $e > 1$ the total revenue curve has a positive slope, that is, it is still increasing, and hence has not reached its maximum point given that
 $P > 0$ and $(1 - 1/e) > 0$; hence $MR > 0$
- If $e < 1$ the total revenue curve has a negative slope, that is, it is falling, given
 $P > 0$ and $(1 - 1/e) < 0$; hence $MR < 0$

We may summarize these results as follows:

- If the demand is inelastic ($e < 1$), an increase in price leads to an increase in total revenue, and a decrease in price leads to a fall in total revenue
- If the demand is elastic ($e > 1$), an increase in price will result in a decrease in total revenue
- If the demand has unitary elasticity, total revenue is not affected by changes in price, since if $e = 1$, then $MR = 0$.

The Price Elasticity of Demand and Its Determinants

- Availability of Close Substitutes
- Necessities versus Luxuries
- Definition of the Market
- Time Horizon

The Price Elasticity of Demand and Its Determinants

Availability of Close Substitutes

- *Goods with close substitutes tend to have more elastic demand because it is easier for consumers to switch from that good to others.*
- *For example, butter and margarine are easily substitutable.*
- *A small increase in the price of butter, assuming the price of margarine is held fixed, causes the quantity of butter sold to fall by a large amount.*

The Price Elasticity of Demand and Its Determinants

Necessities versus Luxuries

- *Necessities tend to have inelastic demands, whereas luxuries have elastic demands.*
- *When the price of a doctor's visit rises, people do not dramatically reduce the number of times they go to the doctor, although they might go somewhat less often.*

The Price Elasticity of Demand and Its Determinants

Definition of the Market

- *The elasticity of demand in any market depends on how we draw the boundaries of the market.*
- *Narrowly defined markets tend to have more elastic demand than broadly defined markets because it is easier to find close substitutes for narrowly defined goods.*
- *For example, food, a broad category, has a fairly inelastic demand because there are no good substitutes for food.*
- *Vanilla ice cream, a very narrow category, has a very elastic demand because other flavors of ice cream are almost perfect substitutes for vanilla.*

The Price Elasticity of Demand and Its Determinants

Time Horizon

- *Goods tend to have more elastic demand over longer time horizons. When the price of gasoline rises, the quantity of gasoline demanded falls only slightly in the first few months.*
- *Over time, however, people buy more fuel-efficient cars, switch to public transportation, and move closer to where they work.*
- *Within several years, the quantity of gasoline demanded falls more substantially.*

The Price Elasticity of Demand and Its Determinants

- Demand tends to be more elastic :
 - the larger the number of close substitutes.
 - if the good is a luxury.
 - the more narrowly defined the market.
 - the longer the time period.

Income Elasticity of Demand

- *Income elasticity of demand* measures how much the quantity demanded of a good responds to a change in consumers' income.
- It is computed as the percentage change in the quantity demanded divided by the percentage change in income.

Computing Income Elasticity

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

Income Elasticity

- Types of Goods
 - Normal Goods
 - Inferior Goods
- Higher income raises the quantity demanded for normal goods but lowers the quantity demanded for inferior goods.

Income Elasticity

- Goods consumers regard as necessities tend to be income inelastic
 - Examples include food, fuel, clothing, utilities, and medical services.
- Goods consumers regard as luxuries tend to be income elastic.
 - Examples include sports cars, furs, and expensive foods.

The Cross-Price Elasticity of Demand

- The **cross-price elasticity of demand** measures how the quantity demanded of one good responds to a change in the price of another good.
- It is calculated as the percentage change in quantity demanded of good 1 divided by the percentage change in the price of good 2. That is

$$\text{cross-price elasticity of demand} = \frac{\text{percentage change in quantity demanded of good 1}}{\text{percentage change in the price of good 2}}$$

Cross Price Elasticity of Demand

- *The responsiveness of demand of one good to changes in the price of a related good – either a substitute or a complement*

$$E_{X,Y} = \left(\frac{\% \Delta Q_x}{\% \Delta P_y} \right)$$

$$E_{X,Y} = \left(\frac{\Delta Q_x}{\Delta P_y} \right) \left(\frac{P_y}{Q_x} \right)$$

- **Goods which are complements:**
Cross Elasticity will have negative sign (inverse relationship between the two)
- **Goods which are substitutes**
:Cross Elasticity will have a positive sign (positive relationship between the two)
- For unrelated goods Cross Elasticity is zero

Example

- **Example:** Consider the following data which shows the changes in quantity demanded of good X in response to changes in the price of good Y.
- Calculate the cross –price elasticity of demand between the two goods. What can you say about the two goods?

Unit price of Y	Quantity demanded of X
10	1500
15	1000

THE ELASTICITY OF SUPPLY

- *Price elasticity of supply* is a measure of how much the quantity supplied of a good responds to a change in the price of that good.
- Price elasticity of supply is the percentage change in quantity supplied resulting from a percent change in price.

The Elasticity of Supply

- Computing price elasticity of supply
 - Percentage change in quantity supplied divided by percentage change in price
 - Always positive
- Midpoint method
 - Two points: (Q_1, P_1) and (Q_2, P_2)

$$\text{Price elasticity of supply} = \frac{(Q_2 - Q_1) / [(Q_2 + Q_1) / 2]}{(P_2 - P_1) / [(P_2 + P_1) / 2]}$$

Example

- Suppose that an increase in the price of milk from \$2.85 to \$3.15 a gallon raises the amount that dairy farmers produce from 9,000 to 11,000 gallons per month. Using the midpoint method, calculate price elasticity of supply

The Elasticity of Supply

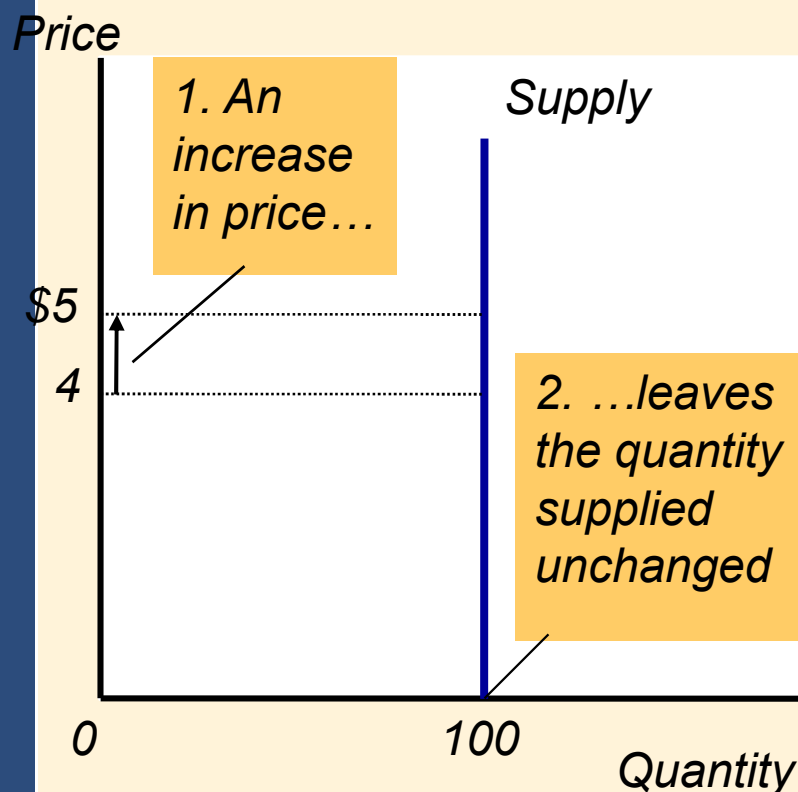
- Variety of supply curves
 - Supply is unit elastic
 - Price elasticity of supply = 1
 - Supply is elastic
 - Price elasticity of supply > 1
 - Supply is inelastic
 - Price elasticity of supply < 1

The Elasticity of Supply

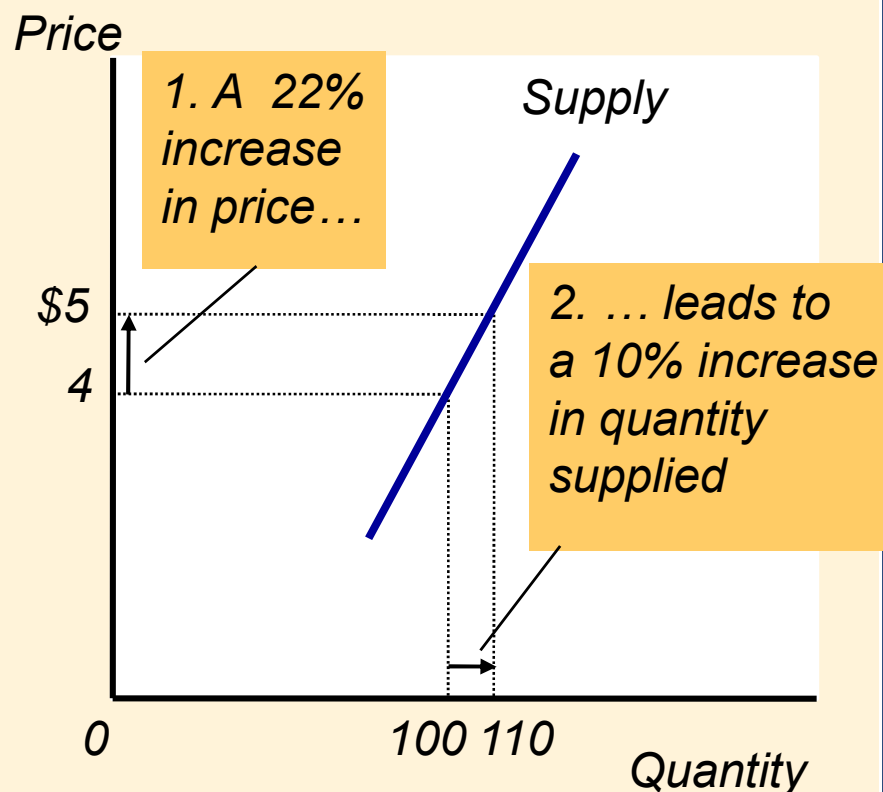
- Variety of supply curves
 - Supply is perfectly inelastic
 - Price elasticity of supply = 0
 - Supply curve – vertical
 - Supply is perfectly elastic
 - Price elasticity of supply = infinity
 - Supply curve – horizontal

The Price Elasticity of Supply (a, b)

(a) Perfectly Inelastic Supply: Elasticity Equals 0



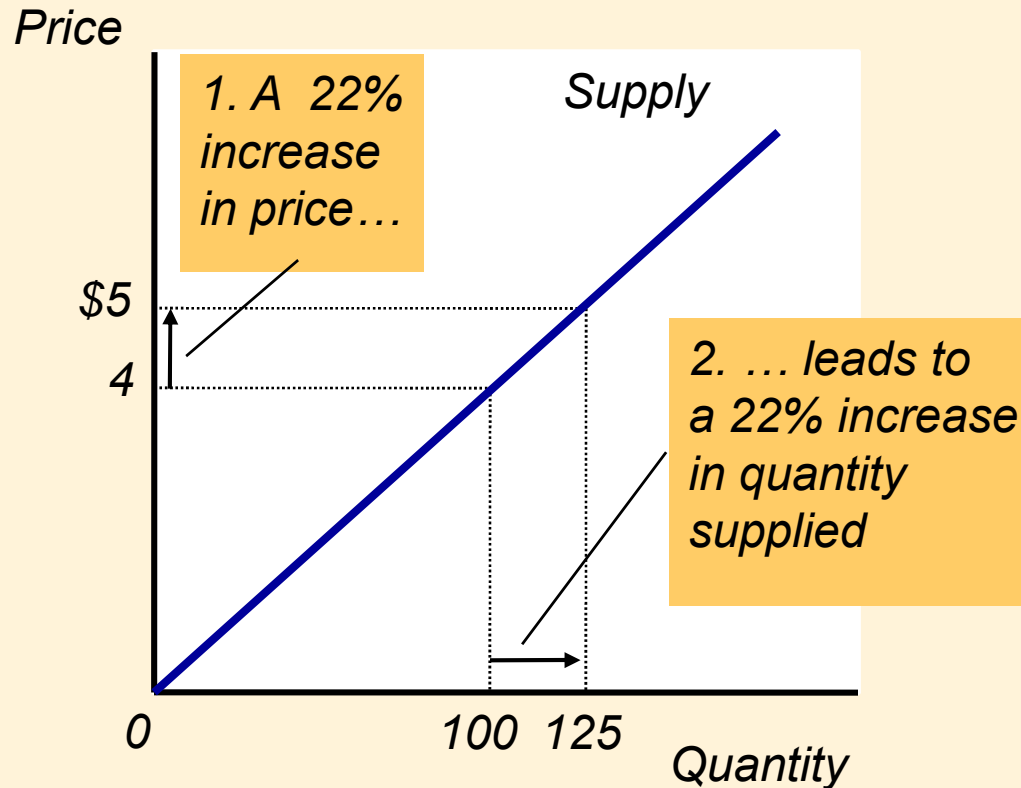
(b) Inelastic Supply: Elasticity Is Less Than 1



The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

The Price Elasticity of Supply (c)

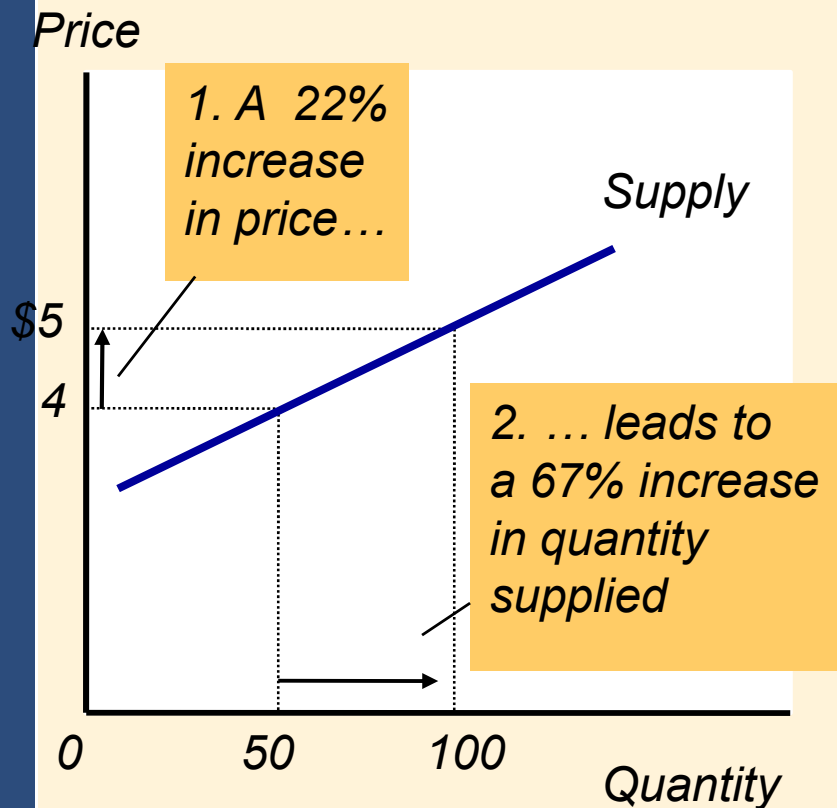
(c) Unit Elastic Supply: Elasticity Equals 1



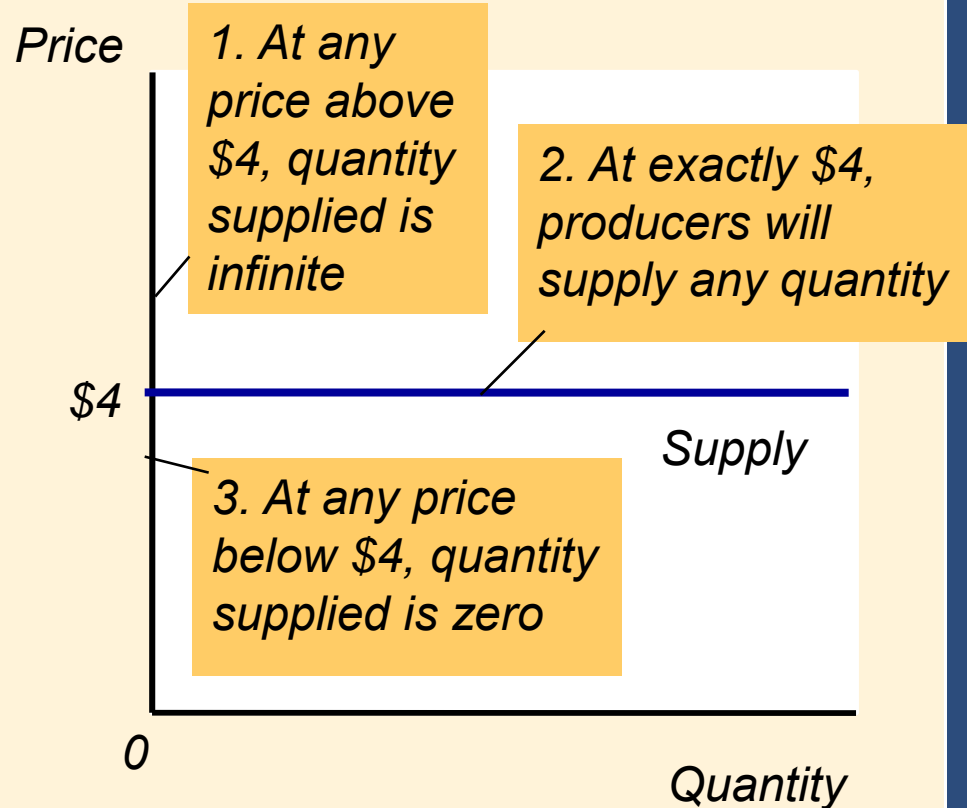
The price elasticity of supply determines whether the supply curve is steep or flat.

The Price Elasticity of Supply (d, e)

(d) Elastic Supply: Elasticity Is Greater Than 1



(e) Perfectly Elastic Supply: Elasticity Equals Infinity



The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

Determinants of Elasticity of Supply

- Ability of sellers to change the amount of the good they produce.
 - Beach-front land is inelastic.
 - Books, cars, or manufactured goods are elastic.
- Time period.
 - Supply is more elastic in the long run.

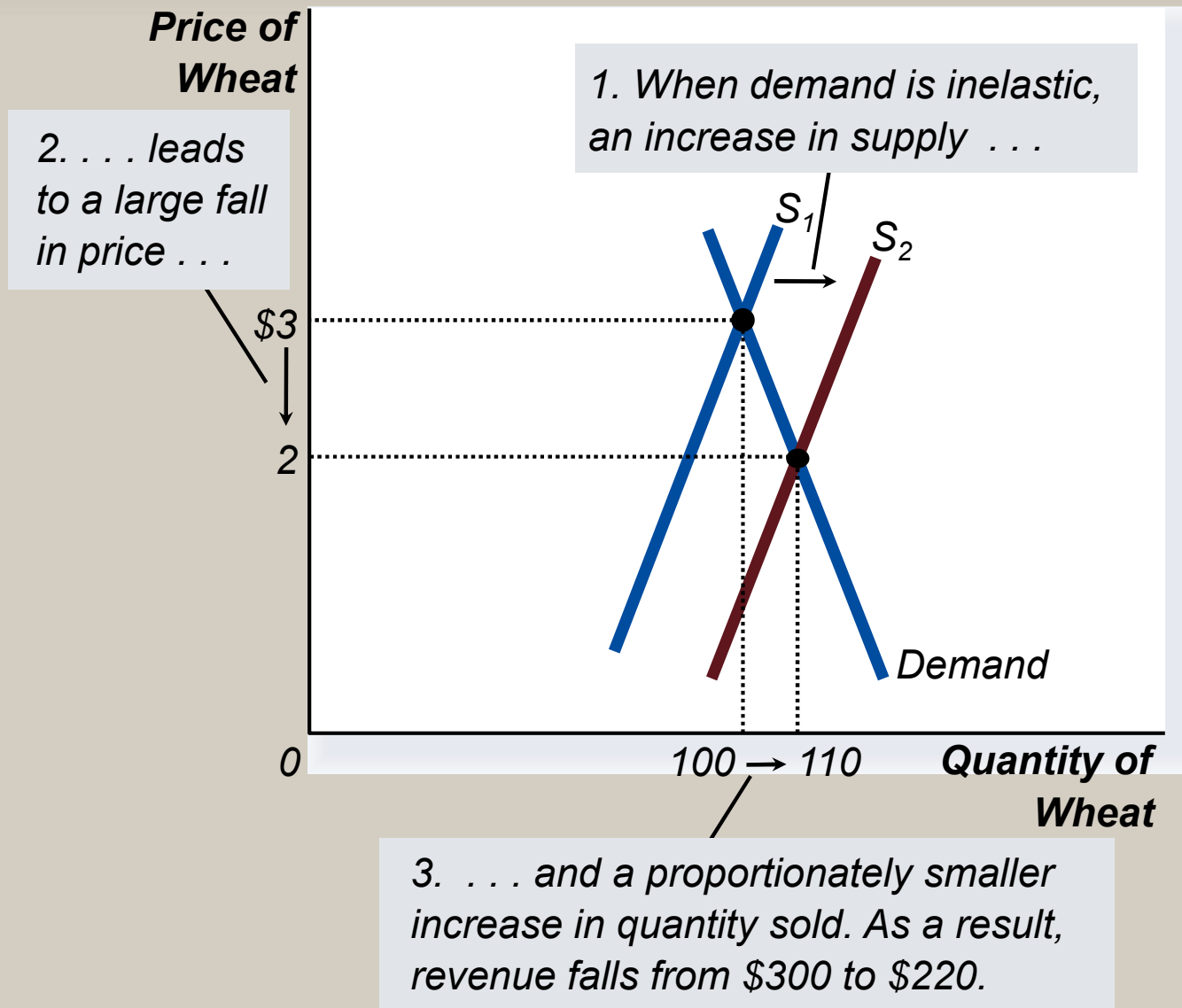
APPLICATION of ELASTICITY

- Can good news for farming be bad news for farmers?
- What happens to wheat farmers and the market for wheat when university agronomists discover a new wheat hybrid that is more productive than existing varieties?

THE APPLICATION OF SUPPLY, DEMAND, AND ELASTICITY

- Examine whether the supply or demand curve shifts.
- Determine the direction of the shift of the curve.
- Use the supply-and-demand diagram to see how the market equilibrium changes.

Figure 8 An Increase in Supply in the Market for Wheat



Cont....

- If farmers are made worse off by the discovery of this new hybrid, one might wonder why they adopt it.
- The answer goes to the heart of how competitive markets work. Because each farmer is only a small part of the market for wheat, she takes the price of wheat as given.
- For any given price of wheat, it is better to use the new hybrid to produce and sell more wheat.
- Yet when all farmers do this, the supply of wheat increases, the price falls, and farmers are worse off.

Quiz

- *How might a drought that destroys half of all farm crops be good for farmers?*
- *If such a drought is good for farmers, why don't farmers destroy their own crops in the absence of a drought?*

Summary

- Price elasticity of demand measures how much the quantity demanded responds to changes in the price.
- Price elasticity of demand is calculated as the percentage change in quantity demanded divided by the percentage change in price.
- If a demand curve is elastic, total revenue falls when the price rises.
- If it is inelastic, total revenue rises as the price rises.

Summary

- The income elasticity of demand measures how much the quantity demanded responds to changes in consumers' income.
- The cross-price elasticity of demand measures how much the quantity demanded of one good responds to the price of another good.
- The price elasticity of supply measures how much the quantity supplied responds to changes in the price. .

Summary

- In most markets, supply is more elastic in the long run than in the short run.
- The price elasticity of supply is calculated as the percentage change in quantity supplied divided by the percentage change in price.
- The tools of supply and demand can be applied in many different types of markets.

Exercise 1

Given market demand $Q_d = 50 - P$, and market supply $P = Q_s + 5$

- a) Find the market equilibrium price and quantity?
- b) What would be the state of the market if market price was fixed at Birr 25 per unit?
- c) Calculate and interpret price elasticity of demand at the equilibrium point.

Exercise 2

- When price of tea in local café rises from Br. 10 to 15 per cup, demand for coffee rises from 3000 cups to 5000 cups a day despite no change in coffee prices.
 - A) Determine cross price elasticity.
 - B) Based on the result, what kind of relation exists between the two goods?